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FACULTY OF COMPUTING

SECD2613 SYSTEM ANALYSIS AND DESIGN
SECTION 02

SEMESTER 2 2025/2026

TOPIC:

PHASE 3

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1.0 Overview of the Project

The **Autism Early Screening Support System** and **Daily Routine Management System** is a proposed web-based platform designed to help parents and NGOs to improve autism support through early screening and a structured daily routine management. The system addresses three major issues which are delayed autism diagnosis and lack of autism-specific routine management tools. It integrates an M-CHAT-R-based autism screening module with automated result generation, reporting features and a daily routine management module that supports feeding, sleep and hygiene schedules through visual reminders, countdown timers, and sensory friendly interfaces. The project was developed following the planning, analysis, and design phases of the System Development Life Cycle (SDLC), with requirements gathered through questionnaires and interviews. Findings revealed that the current practices rely heavily on manual methods such as memory resulting in inconsistent tracking and difficulty of sharing information. The proposed system aims to provide a centralized, accessible, and user-friendly solution that improves early autism awareness, supports routine consistency, and contributes to better developmental outcomes and quality for individuals with Autism Spectrum Disorder.

2.0 Problem Statement

Problem 1 - Late or Missed Early Diagnosis of Autism Spectrum Disorder (ASD)
Parents currently lack access to a structured and reliable platform for identifying early signs of Autism Spectrum Disorder (ASD). Early concerns are often based on informal observations resulting in uncertainties and delays in seeking a professional assessment. The absence of an accessible screening approach contributes to delayed diagnosis.

Problem 2 - Lack of Structured Daily Routine Support for Individuals with ASD
Individuals with ASD often experience difficulties in managing daily activities such as feeding, sleeping, hygiene, and personal routines. The existing applications do not sufficiently address the specific needs of individuals with ASD. As a result, parents face challenges in maintaining consistent routines that support independence and daily functioning.

Problem 3 - Scattered Communication and Data Management
The current ASD support practices that involve channels for screening information and daily routine monitoring. The lack of a centralized platform causes the scatter in information sharing and actually will limit the availability of comprehensive information required for effective assessment, progress monitoring, and intervention planning.

3.0 Proposed Solutions

To address the identified problems, this project proposed the design and development of an integrated web-based platform that combines early screening, daily routine management, and healthcare collaboration features within a single platform. The system is designed with two main components, the Autism Early Screening Module and the Autism Daily Routine Management Module.

The **Autism Early Screening Module** is designed to assist parents in identifying the early signs of ASD through a structured digital questionnaire. The module also provides a user-friendly interface that guides parents through the screening process, stores assessment records, and generates screening summaries that can support further consultation with healthcare providers.

The **Autism Daily Routine Management Module** is designed to support individuals with ASD in developing consistent daily habits through personalized visual schedules and task management features. This module allows parents to create, customize, and monitor routines related to feeding, sleep, hygiene, and other daily activities. The design incorporates visual communication elements, reminders, and progress tracking to accommodate different behavioural and sensory needs.

The overall system design focuses on usability, accessibility, and effective information management by integrating screening and routine support into one platform. Through this integrated approach, the proposed system aims to improve early ASD identification and encourage independence in daily activities.

4.0 Current Business Process/Workflow

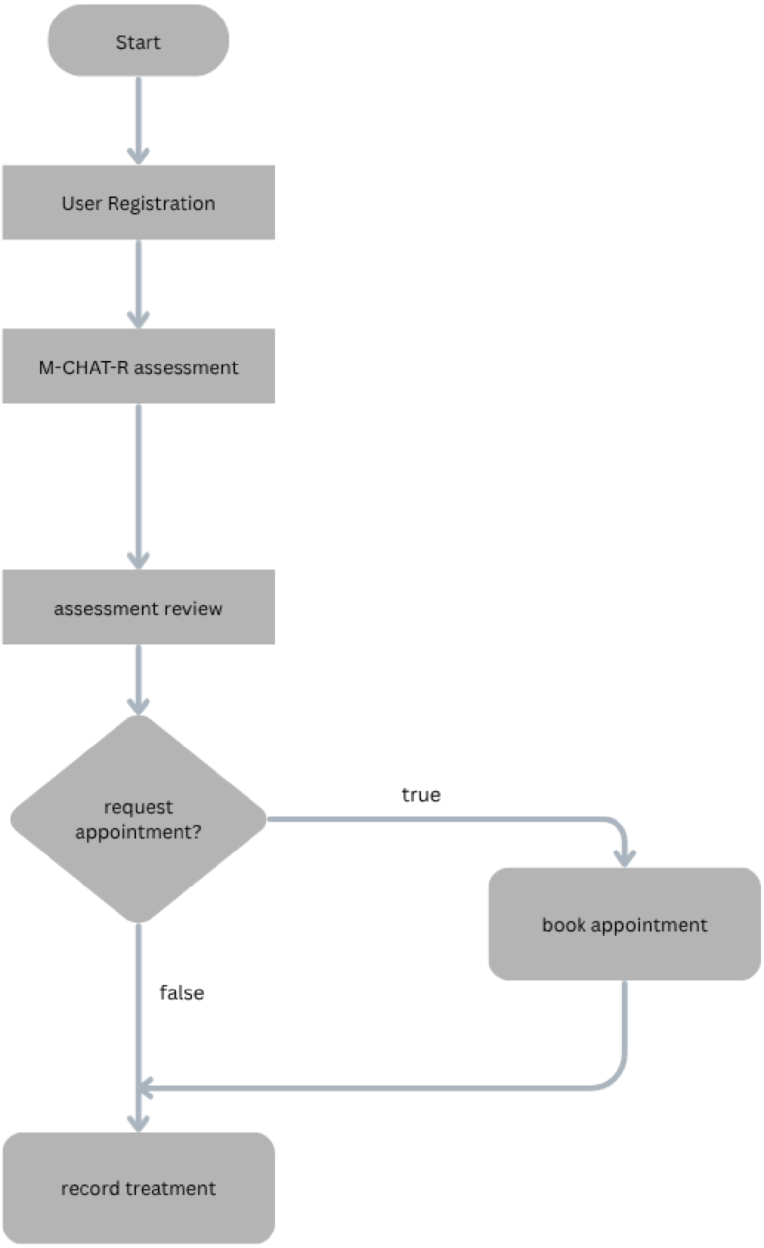

Figure 1: Figure shows the process flow of an autism assessment and treatment management system

The process begins when a user registers. After a successful registration process, the user can proceed to complete the M-CHAT-R assessment, which is used to evaluate the child's autism risk level. Once the assessment is completed, the results are reviewed during the assessment review stage. Then, the user will reach the first decision point where the user can choose to request an appointment. If the user selects true, they will proceed to book an appointment and proceed with a record treatment stage simultaneously. If the user selects false, the process will directly proceed with the recording treatment stage. This flow ensures that users can complete screening, receive assessment results, and schedule professional support when needed.

5.0 Logical DFD (AS-IS)

5.1 Context Diagram

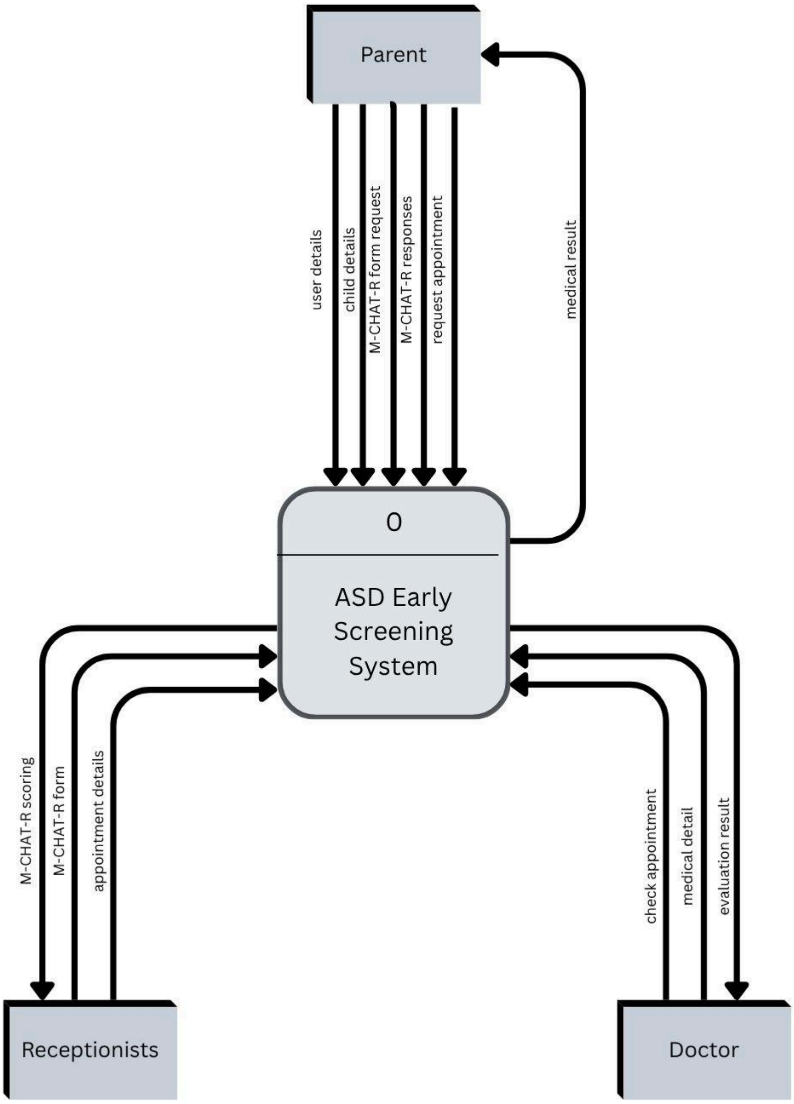

Figure 2: Figure shows the context diagram of the Autism Spectrum Disorder (ASD) Early Screening System

This figure shows the interactions between the system and its external users. The system receives user details, child details, M-CHAT-R requests, responses, and appointment requests from parents. It also communicates with receptionists by receiving M-CHAT-R scoring information, appointment details, and updates. Doctors interact with the system by accessing appointment information, medical details, and providing evaluation results. After processing the collected information, the system returns relevant outputs such as medical results, screening outcomes, and appointment-related information to the respective users. Overall, the diagram illustrates how data flows between the parent, receptionist, doctor, and the ASD Early Screening System.

5.2 Diagram 0

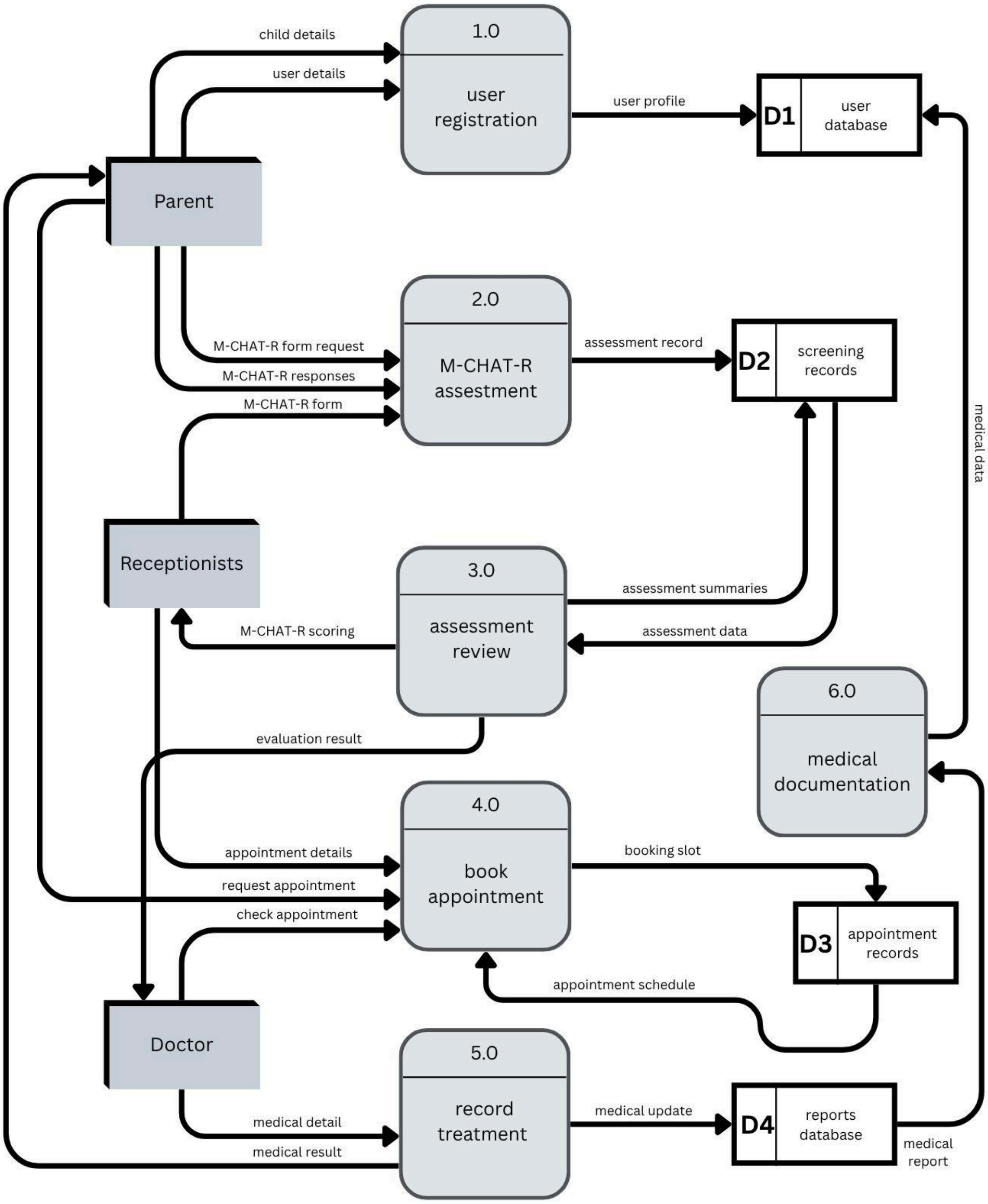

Figure 3: Figure shows the level 1 diagram of the Autism Spectrum Disorder (ASD) Early Screening System

This figure shows the main processes, external users, and data storage involved in the system. The system consists of six main processes which are the user registration, M-CHAT-R assessment, assessment review, appointment booking, treatment recording, and medical documentation. Parents provide user and child details during registration, complete the M-CHAT-R assessment, and request appointments. Receptionists manage screening scores and appointment information, while doctors provide medical details, evaluation results, and treatment updates. The system stores and retrieves information through databases such as the user database, screening records, appointment records, and reports database. These processes allow the system to manage screening results, schedule appointments, document medical information, and generate reports efficiently for parents, receptionists, and doctors.

6.0 System Analysis and Specification

6.1 Logical DFD TO-BE System

6.1.1 Context Diagram

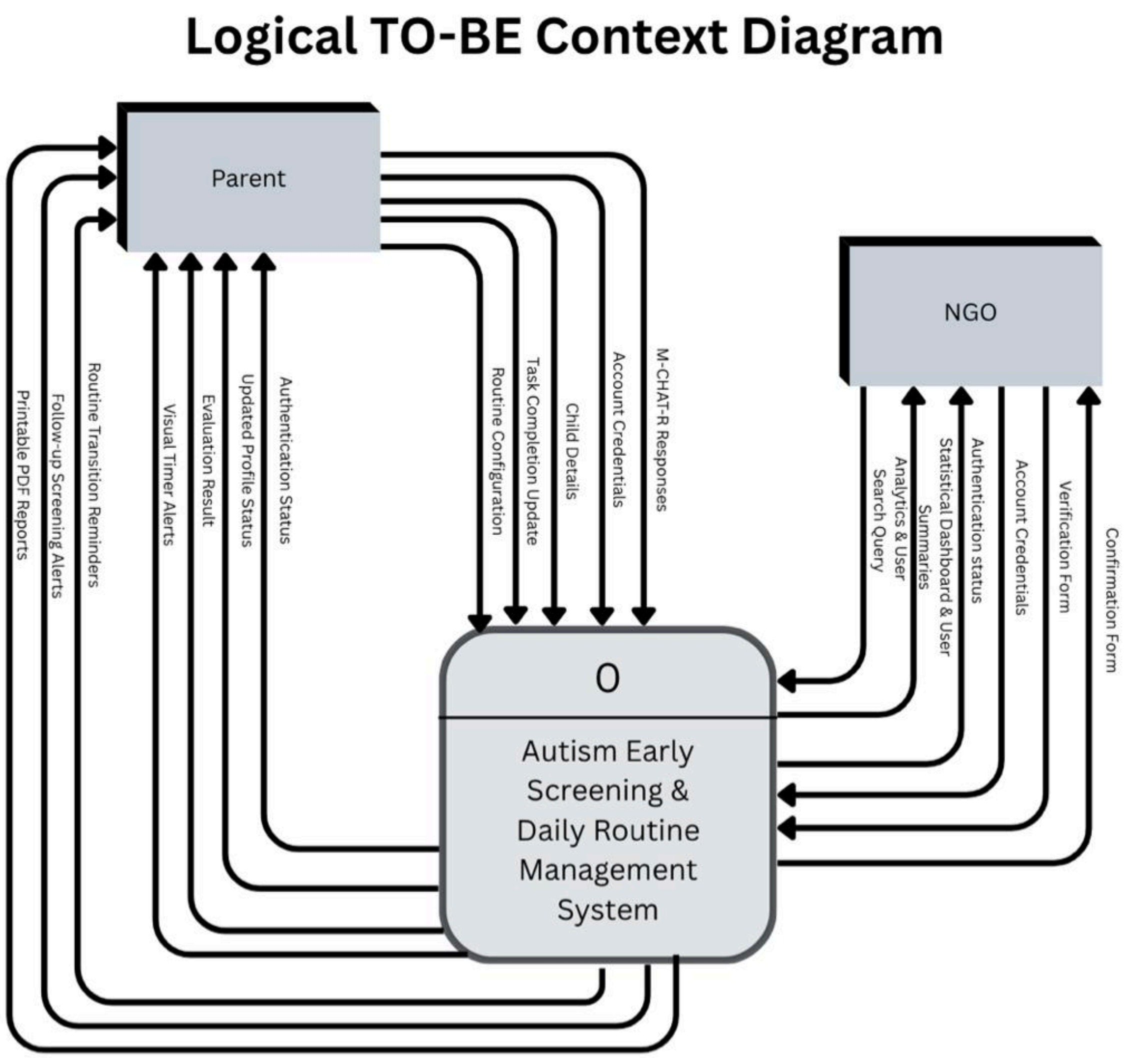

Figure 4: Figure show the entire Autism Early Screening & Daily Routine Management System

This diagram illustrates the core system as a single central process interacting with two main external entities: the Parent and the NGO. Parents input data like child details, M-CHAT-R responses, and routine updates, while receiving alerts, evaluation results, and printable reports. Meanwhile, the NGO interacts with the system to verify profiles and submit analytical search queries, receiving statistical dashboards and confirmation forms in return.

6.1.2 Diagram 0

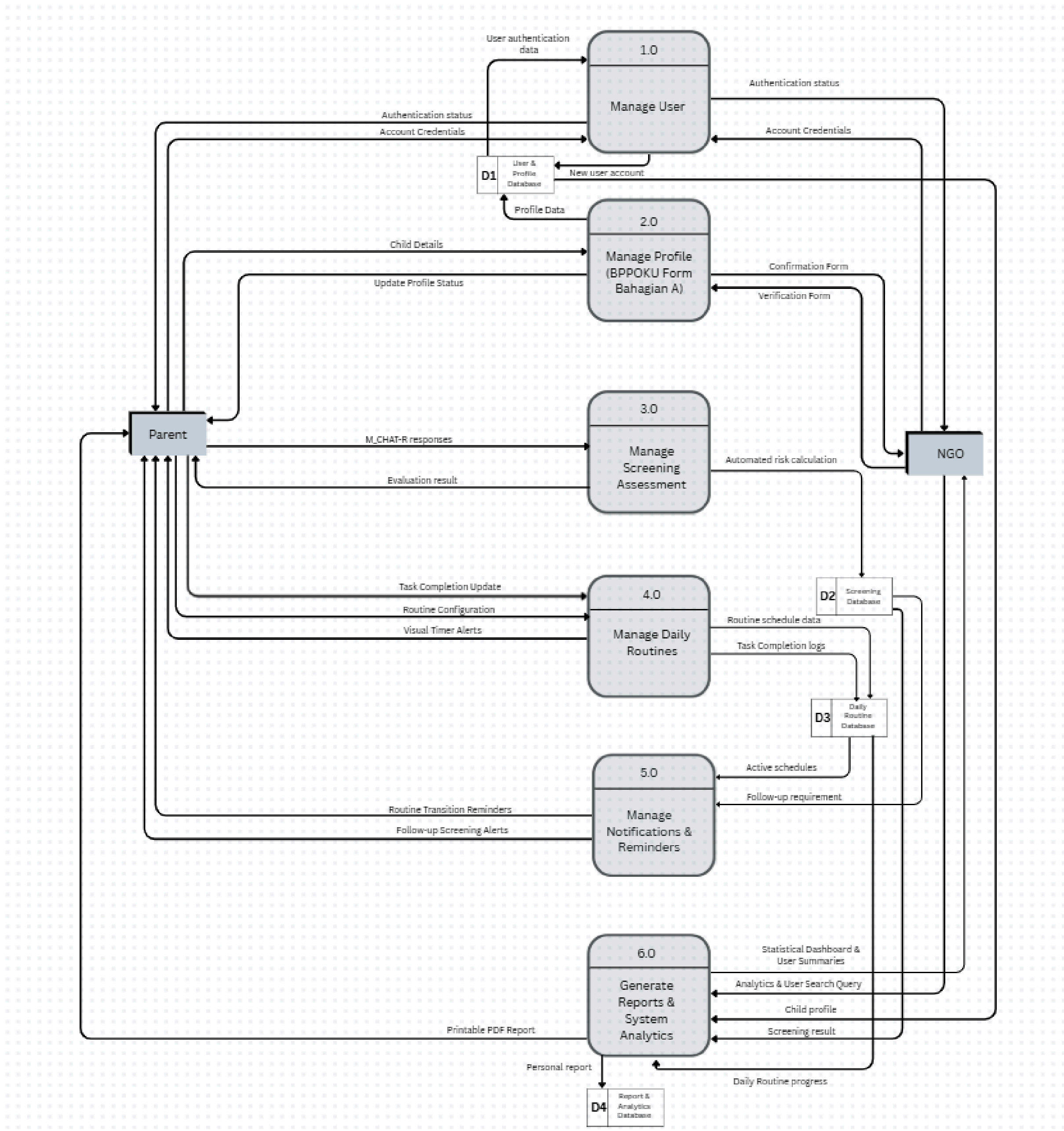

Figure 5: Figure shows the breaks of the main system down into its six primary logical processes

Diagram 0 shows how our system actually works. It maps out exactly how data flows from Parent and NGO through user management (1.0), profile management (2.0), screening assessments (3.0), daily routines (4.0), notifications and reminders (5.0), and analytics generation (6.0). This diagram also introduces the internal data stores (D1 to D4) to clearly illustrate where profile, screening, routine, and analytics data are logically held at rest within the system.

Link for clearer picture: https://c-anva.link/s5jcmrlcm81sz2x

6.1.3 Child Diagram

- Manage User

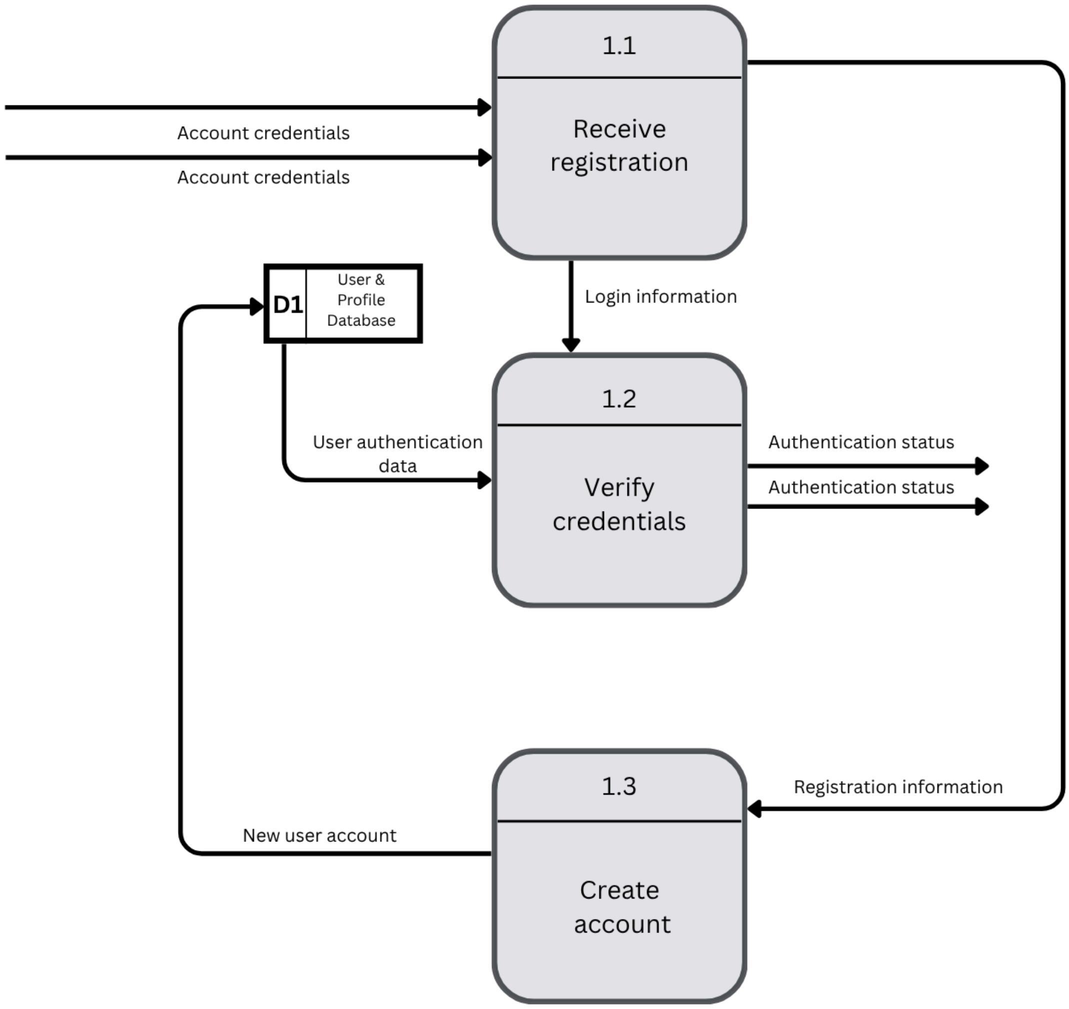

Figure 6: Figure shows the child diagram of manage user (1.0)

This child diagram details the internal logic of the system's security and access control. It explains how login and registration credentials from both Parents and NGOs are received, validated against the User & Profile Database (D1), and processed to either generate a new account or return an authentication status granting access.

- Manage Profile (BPPOKU Form)

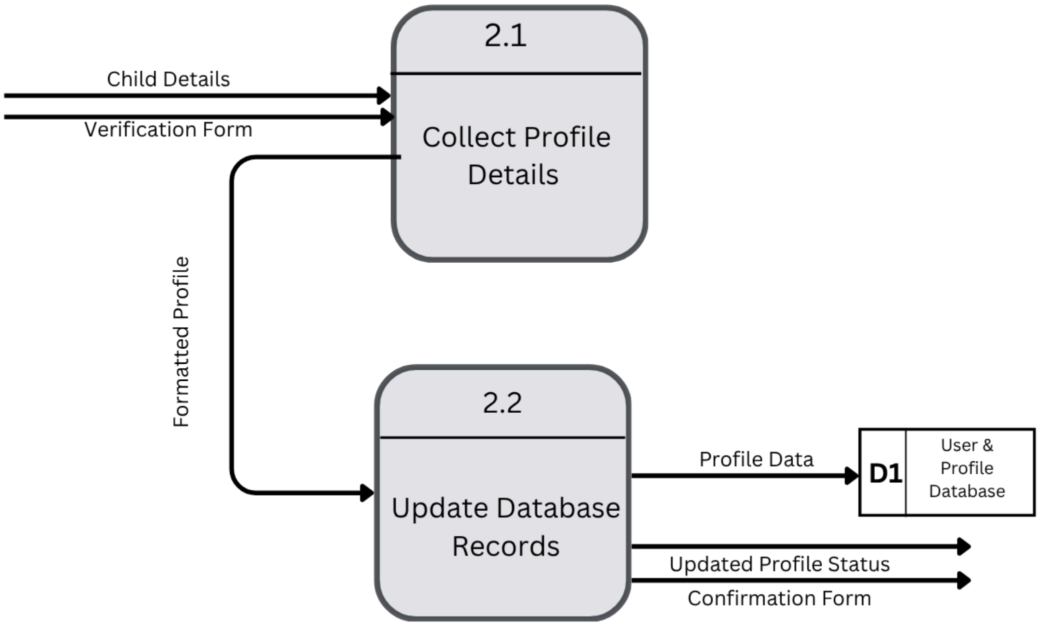

Figure 7: Figure shows the child diagram of manage profile (2.0)

This figure shows the logical flow of a Parent submitting child details via the BPPOKU Form (Bahagian A), the system saving it to the database, and the NGO reviewing this submitted data to issue a formal verification status.

- Manage Screening Assessment

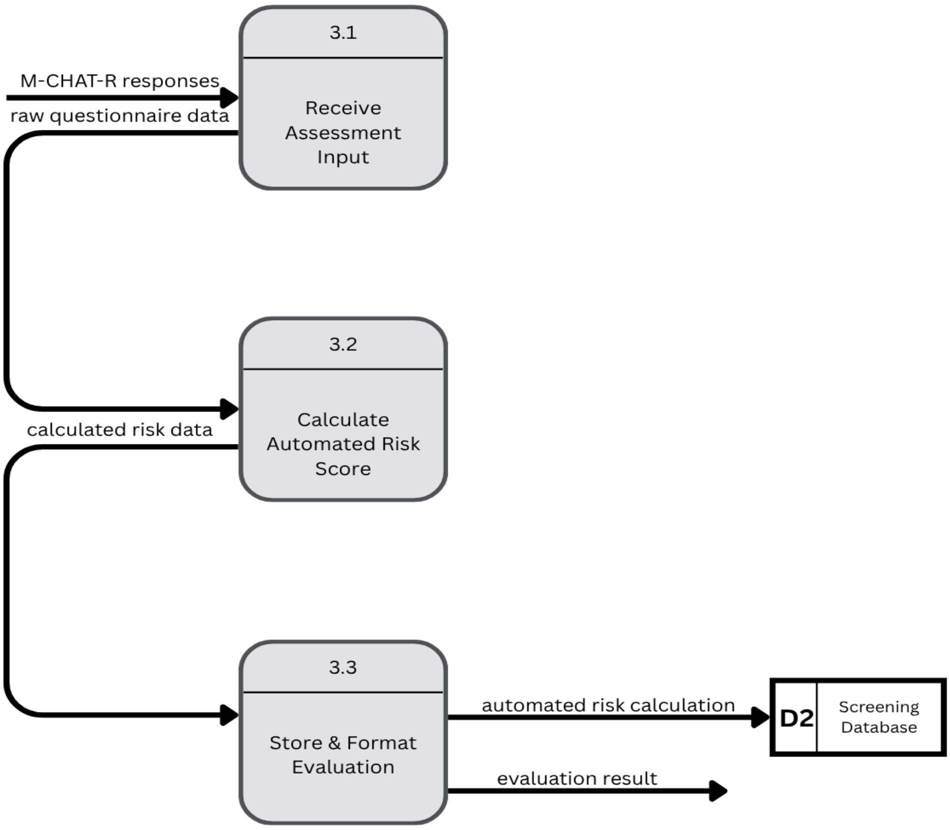

Figure 8: Figure shows the child diagram of manage screening assessment (3.0)

This diagram breaks down the logic behind the M-CHAT-R diagnostic tool. It illustrates how the system ingests the parent's quiz responses, runs an automated risk calculation algorithm to process the answers, and outputs the final evaluation result so it can be displayed to the parent and saved to the Screening Database (D2).

- Manage Daily Routines

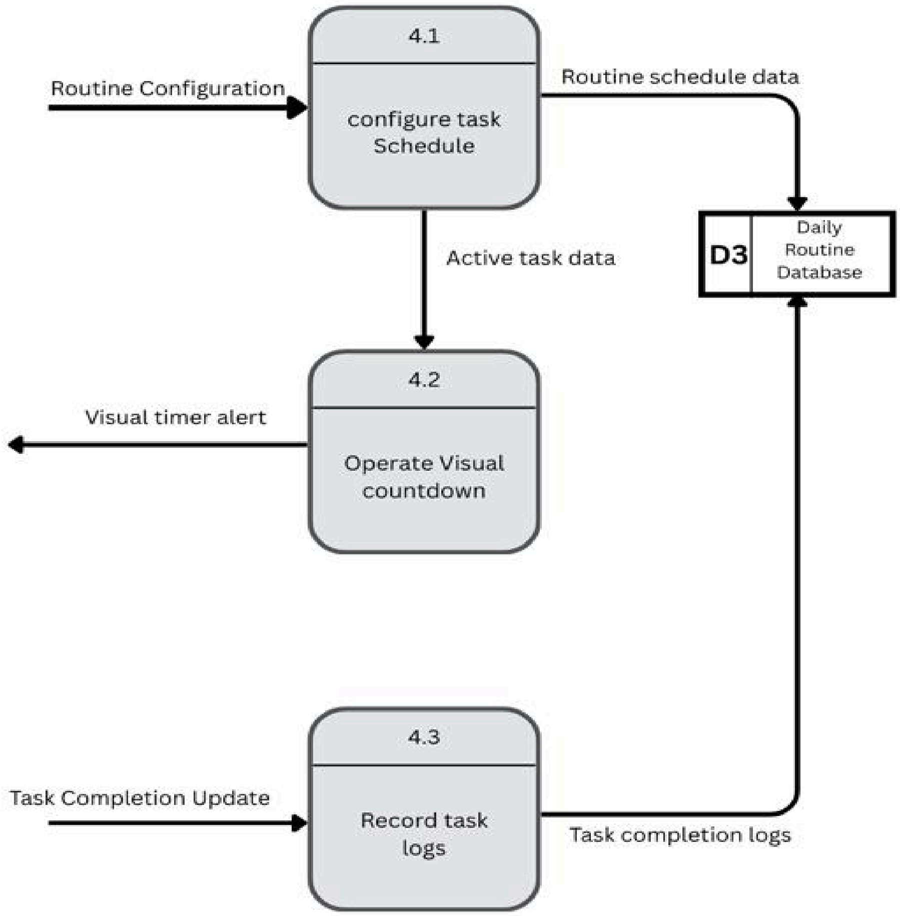

Figure 9: Figure shows the child diagram of manage daily routine (4.0)

The figure explains the logic of Parents configuring routine schedules, the system executing visual timer alerts, and the subsequent tracking of task completion logs, all of which are continuously stored within the Daily Routine Database (D3).

- Manage Notifications & Reminders

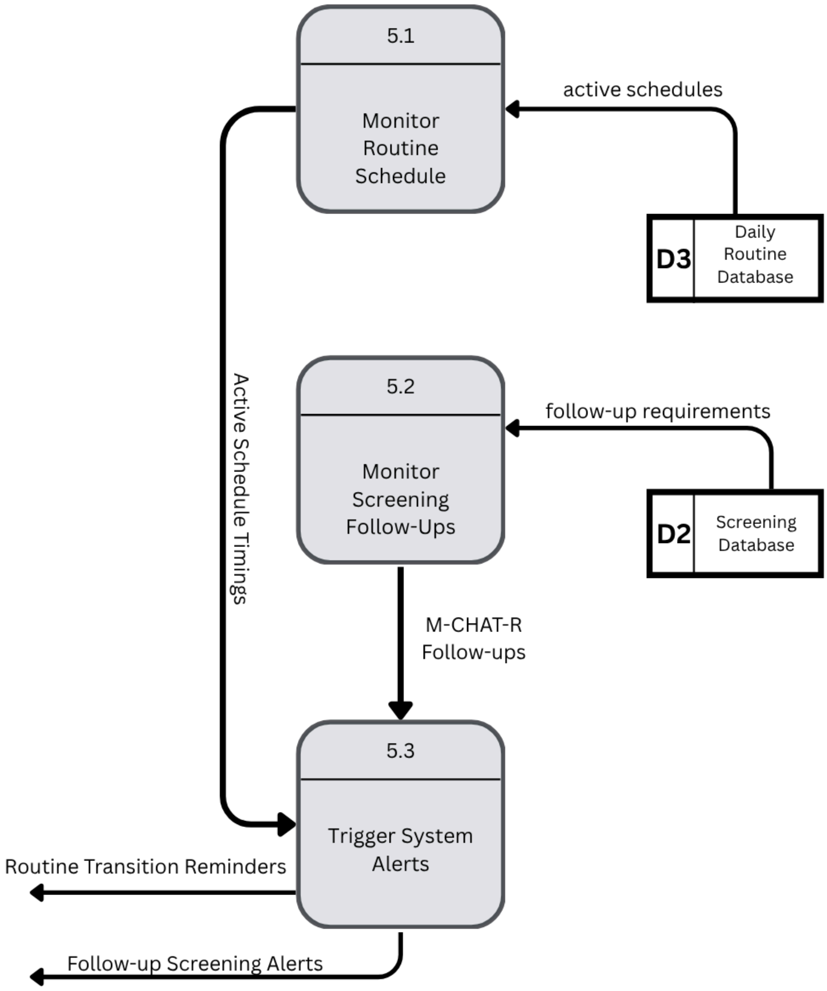

Figure 10: Figure shows the child diagram of manage notifications and reminders (5.0)

This diagram details the system's automated, time-based background processes. It maps out how the system continuously scans both the Daily Routine Database (D3) and the Screening Database (D2) to automatically trigger and push routine transition reminders and follow-up screening alerts to the Parent at exactly the right time.

- Generate Reports & System Analytics

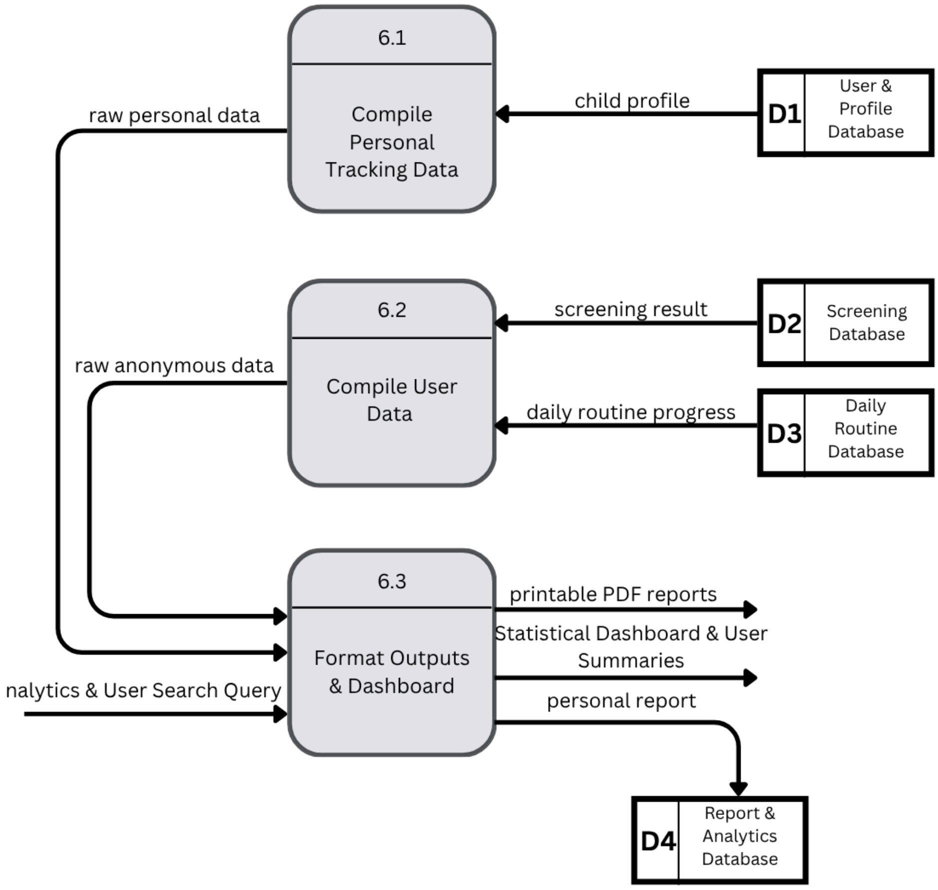

Figure 11: Figure shows the child diagram of generate reports and system analytics (6.0)

This diagram explains the system's data aggregation capabilities. It shows how raw data is pulled from the system's various databases to compile two distinct outputs which is a personalized, printable PDF report requested by the Parent, and a searchable, aggregated statistical dashboard queried by the NGO.

6.2 Process Specification (based on Logical DFD TO-BE)

Table 1: Process Specification Process 1.1

<table>
<tr><th colspan="2">Process Specification : Process 1.1</th></tr>
<tr><td>Number</td><td>1.1</td></tr>
<tr><td>Name</td><td>Receive registration</td></tr>
<tr><td>Description</td><td>Receive incoming account credentials to route them for identity check or account initialization.</td></tr>
<tr><td>Input Data Flow</td><td>Account Credentials (from Parents or NGO)</td></tr>
<tr><td>Output Data Flow</td><td>Login Information (to Process 1.2)
Registration Information (to Process 1.3)</td></tr>
<tr><td>Type of Process</td><td>☑ Online
☐ Batch
☐ Manual</td></tr>
<tr><td>Process Logic</td><td>IF Account Credentials are submitted THEN
    Extract Account Credentials into Login information data flow
    Forward Login information to Process 1.2
    Extract Account Credentials into Registration information data flow
    Forward Registration information to Process 1.3
ELSE
    Display error “Credentials cannot be empty”
ENDIF</td></tr>
<tr><td>Refer to Name</td><td>☐ Structured English
☐ Decision Table
☐ Decision Tree</td></tr>
<tr><td>Unresolved Issues</td><td>None</td></tr>
</table>

Table 1 presents the initial system execution requirements for Process 1.1. The process is categorized as an Online process because it requires direct interaction to receive account credentials during registration or login. The main function of this process is to receive and organize input data. If the received data is valid and not empty, the system separates it into login information and registration information. The login information is forwarded to the login verification process, while the registration information is sent to the registration process for further validation. Since the process only involves simple data routing and validation, no additional Structured English, Decision Table or Decision Tree is required. No unresolved issues have been identified for this process

Table 2: Process Specification Process 1.2

Process Specification : Process 1.2	
Number	1.2
Name	Verify credentials
Description	Validates user login information and authentication data against existing accounts stored in the database to prevent duplication and issue an authentication status.
Input Data Flow	Login Information (from Process 1.1) User Authentication Data (from D1: User & Profile Database)
Output Data Flow	Authentication Status (to Parents or NGO)
Type of Process	☑ Online ☐ Batch ☐ Manual

Process Logic	READ User authentication data FROM D1: User & Profile Database
	IF Login information matches existing records THEN
	    Set Authentication status = "Access Granted"
	ELSE
	    Set Authentication status = "Access Denied"
	ENDIF
	OUTPUT Authentication status to user interface

Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 2 shows the operational details of Process 1.2. This online process verifies the user's login credentials by comparing the input data with the authentication records stored in database D1. The system then produces a validation status based on the matching result. As the process follows a straightforward validation procedure, no additional Structured English, Decision Table or Decision Tree. No unresolved issues have been identified for this process.

Table 3: Process Specification Process 1.3

Process Specification : Process 1.3	
Number	1.3
Name	Create Account
Description	Commit the verified registration information into the profile database to create a new user account profile within the system.
Input Data Flow	Registration Information (from process 1.1)
Output Data Flow	New User Account (to D1: User & Profile Database)
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	RECEIVE Registration information Format structural elements of Registration information into database format WRITE New user account record into D1: User & Profile Database
Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 3 describes the specifications for Process 1.3. The online process handles user registration by receiving validated registration information and organizing it into the required account structure. The system stores the new user account information in the User & Profile Database (D1). There is no additional Structured English, Decision Table or Decision Tree required and no unresolved issues for this process.

Table 4: Process Specification Process 2.1

Process Specification : Process 2.1	
Number	2.1
Name	Collect Profile Details
Description	Combines child details from the parents and the verification form from the NGO into a single formatted profile.
Input Data Flow	Child Details (from Parent) Verification Form (from NGO)
Output Data Flow	Formatted Profile (to Process 2.2)
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	RECEIVE Child Details AND Verification Form Validate that mandatory data attributes are complete Compile attributes together into Formatted Profile Forward Formatted Profile to Process 2.2
Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 4 outlines the process logic of Process 2.1. As an online process, it received child details from the parents and verification form information from the NGO. The system combines these inputs to create a formatted profile for further processing. Due to the straightforward data collection, no additional Structured English, Decision Table or Decision Tree required and no unresolved issues for this process.

Table 5: Process Specification Process 2.2

Process Specification : Process 2.2	
Number	2.2
Name	Update Database Records
Description	Updates the child's profile in the database and provides an updated profile status to the parents and confirmation form to the NGOs.
Input Data Flow	Formatted Profile (from Process 2.1)
Output Data Flow	Profile Data (to D1: User & Profile Database) Updated Profile Status Confirmation Form
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	RECEIVE Formatted Profile Extract data elements into Profile Data WRITE Profile Data to D1: User & Profile Database GENERATE Updated Profile Status GENERATE Confirmation Form OUTPUT Updated Profile Status to parents OUTPUT Confirmation Form to NGO
Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 5 describes the process specifications for Process 2.2. This online process updates the information stored in database D1. After the database has been successfully updated, the system sends an updated status to the Parents and generates a confirmation form for the NGO. Since the process only involves database updating, no additional Structured English, Decision Table or Decision Tree required. There are no unresolved issues for this process.

Table 6: Process Specification Process 3.1

Process Specification : Process 3.1	
Number	3.1
Name	Receive Assessment Input
Description	Receive the incoming M-CHAT-R responses submitted by the parents and prepare them for the scoring.
Input Data Flow	M-CHAT-R Responses
Output Data Flow	Raw Questionnaire Data (to Process 3.2)
Type of Process	☑ Online ☐ Batch ☐ Manual

Process Logic	RECEIVE M-CHAT-R responses
	Validate that all required screening questions are answered
	IF validation passes THEN
	Forward raw questionnaire data to Process 3.2
	ELSE
	Display error "Please complete all questions before submitting"
	ENDIF

Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 6 presents the process logic of Process 3.1. This online process receives M-CHAT-R responses submitted by parents and checks whether aa required questions have been completed. Once the responses are verified, the system converts them into questionnaire data for further processing. There is no additional Structured English, Decision Table or Decision Tree required. No unresolved issues have been identified for this process.

Table 7: Process Specification Process 3.2

Process Specification : Process 3.2	
Number	3.2
Name	Calculate Automated Risk Score
Description	Processes the validated raw questionnaire data using the automated M-CHAT-R scoring logic to determine the risk level.
Input Data Flow	Raw Questionnaire Data (from Process 3.1)
Output Data Flow	Calculated Risk Data (to Process 3.3)
Type of Process	☑ Online ☐ Batch ☐ Manual

Process Logic	RECEIVE raw questionnaire data
	Initialize TotalRiskScore = 0
	FOR EACH response IN questionnaire
	    IF response matches the autism criteria for M-CHAT-R THEN
	        Increment TotalRiskScore by 1
	    ENDIF
	ENDFOR
	Package TotalRiskScore into calculated risk data
	Forward calculated risk data to Process 3.3

Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 7 shows the process specifications for Process 3.2. As an online process, it evaluates questionnaire responses and calculates the total screening score based on predefined scoring rules. The generated score is then forwarded for the next stage of assessment. Due to the straightforward scoring procedure, no additional Structured English, Decision Table or Decision Tree required. No unresolved issues have been identified in this process.

Table 8: Process Specification Process 3.3

Process Specification : Process 3.3	
Number	3.3
Name	Store & Format Evaluation
Description	Formats the calculated risk data and automatically stores the final screening results inside the Screening Database.
Input Data Flow	Calculated Risk Data (from Process 3.2)
Output Data Flow	Automated Risk Calculation (to D2: Screening Database) Evaluation Result (to Parents)
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	RECEIVE calculated risk data Format data into a structured record containing Score, Risk Category (Low, Medium, High) and Recommendation Message WRITE formatted record as automated risk calculation to D2: Screening Database GENERATE evaluation result OUTPUT evaluation result to parents UI display
Refer to Name	☑ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 8 describes the process specifications for Process 3.3. This online process stores the screening results in the Screening Database (D2) and generates an evaluation result for the parents. Based on the calculated screening score, the system formats result in a clear and understandable evaluation message to the parent interface display. As this process involves specific rules for interpreting scores and generating evaluation results, an additional Structured English is referenced to describe the detailed processing logic. There are no unresolved issues for this process.

Completed Structured English Process 3.3

```
RECEIVE calculated risk data
IF calculated risk data is between 0 and 2 THEN
Set Risk Level = Low Risk
    IF child age is less than 24 months THEN
        Recommend rescreening at 24 months
    ENDIF

ELSEIF calculated risk data is between 3 and 7 THEN
Set Risk Level = Moderate Risk
    Conduct M-CHAT_R Follow-Up Assessment
    IF Follow-Up Score is 2 or higher THEN
        Set Screening Result = Positive
        Refer child for diagnostic evaluation
        Refer child for early intervention services
    ELSE
        Set Screening Result = Negative
        Schedule future routine screening
    ENDIF

ELSE IF total score is between 8 and 20 THEN
Set Risk Level = High Risk
    Skip Follow-Up Assessment
    Refer child directly for diagnostic evaluation
    Refer child for early intervention services
ENDIF

Format data into a structured record containing Score, Risk Category (Low, Medium, High)
and Recommendation Message
WRITE formatted record as automated risk calculation to D2: Screening Database
GENERATE evaluation result
OUTPUT evaluation result to parents UI display
```

Table 9: Process Specification Process 4.1

Process Specification : Process 4.1	
Number	4.1
Name	Configure Task Schedule
Description	Receives and updates daily routine schedules and task information provided by the parents.
Input Data Flow	Routine Configuration
Output Data Flow	Routine Schedule Data (to D3: Daily Routine Database) Active Task Data (to process 4.2)
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	RECEIVE Routine Configurations Format routine details into a schedule structure WRITE Routine Schedule Data TO D3: Daily Routine Database Identify currently active tasks based on system time Forward Active Task Data to process 4.2
Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 9 outlines the process logic of Process 4.1. As an online process, it manages routine updates from the user and stores the updated schedule information in the Daily Routine Database (D3). The system then identifies the currently active task and forwards the active task data to the timer process for further execution. Since the process involves updating routine information and identifying active tasks, no additional Structured English, Decision Table or Decision Tree required and no unresolved issues have been identified.

Table 10: Process Specification Process 4.2

Process Specification : Process 4.2	
Number	4.2
Name	Operate Visual Countdown
Description	Runs the timer subsystem for the active routine task and triggers a visual alert when the timer hits zero.
Input Data Flow	Active Task Data (from process 4.1)
Output Data Flow	Visual Timer Alert (to Parents)
Type of Process	☑ Online ☐ Batch ☐ Manual

Process Logic	RECEIVE Active Task Data
	Extract TaskDuration from data
	Initialize countdown timer using TaskDuration
	WHILE timer is running
	    Update UI Countdown interface
	    IF timer reaches 0 THEN
	        Trigger Visual Timer Alert
	        Exit Loop
	    ENDIF
	ENDWHILE

Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 10 shows the process logic of process 4.2. As an online process, it operates the countdown timer for the currently active task and tracks the remaining duration. Once the timer reaches zero, the system generates visual alerts for the Parents. Due to the simplicity of the countdown and alert process, no additional Structured English, Decision Table or Decision Tree required. No unresolved issues have been identified for this process.

Table 11: Process Specification Process 4.3

Process Specification : Process 4.3	
Number	4.3
Name	Record Task Logs
Description	Records the completion status of a routine task submitted by the parents and stores it as an analytical log record.
Input Data Flow	Task Completion Update
Output Data Flow	Task Completion Logs (to D3: Daily Routine Database)
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	RECEIVE Task Completion Update Generate log record containing Timestamp, Task ID and Completion State (Completed) WRITE Task Completion Logs to D3: Daily Routine Database
Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 11 presents the process specifications for process 4.3. This online process records task completion activities and stores them in the Daily Routine database (D3). When a task status is updated, the system generates a task completion record together with the relevant date and time information. The recorded data is then stored for future tracking and progress monitoring purposes. Since the process only involves recording and storing task completion information, no additional Structured English, Decision Table or Decision Tree required. No unresolved issues have been identified in this process.

Table 12: Process Specification Process 5.1

Process Specification : Process 5.1	
Number	5.1
Name	Monitor Routine Schedule
Description	Regularly checks active schedules in the routine database to track timing limits and identify schedule changes.
Input Data Flow	Active Schedule (from D3: Daily Routine Database)
Output Data Flow	Active Schedule Timings (to Process 5.3)
Type of Process	☑ Online ☐ Batch ☐ Manual

Process Logic	LOOP every 60 seconds
	READ Active Schedules FROM D3: Daily Routine Database
	Compare current system time with task transition thresholds
	IF a match of imminent transition is found THEN
	Forward Active Schedule Timings to Process 5.3
	ENDIF
	ENDLOOP

Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 12 highlights system behaviour within Process 5.1. As an online process, it continuously monitors scheduled data stored in the Daily Routine Database (D3) and compares it with the current system time. Based on the comparison result, the system identifies the active schedule and produces the corresponding timing information. Since the process only involves real-time schedule monitoring and task activation, no additional Structured English, Decision Table or Decision Tree required. There are no unresolved issues for this process.

Table 13: Process Specification Process 5.2

Process Specification : Process 5.2	
Number	5.2
Name	Monitor Screening Follow-Ups
Description	Checks the screening history records in the Screening Database to identify users who are due for an M-CHAT-R re-assessment.
Input Data Flow	Reads Follow-Up Requirements (from D2: Screening Database)
Output Data Flow	M-CHAT-R Follow-ups (to Process 5.3)
Type of Process	☑ Online ☐ Batch ☐ Manual

Process Logic	LOOP daily
	READ Follow-Up requirements FROM D2: Screening Database
	Identify users whose previous M-CHAT-R risk score requires a follow-up assessment after a specific time interval
	IF follow-up criteria are met THEN
	Forward M-CHAT-R Follow-ups data to Process 5.3
	ENDIF
	ENDLOOP

Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 13 outlines the process logic of Process 5.2. This online process monitors screening records in the Screening Database (D2) and compares the reassessment date with the current date. When a follow-up assessment is due, the system generates an M-CHAT-r follow-up notification for the Parent. Due to the straightforward nature of the date-checking and reminder procedure, no additional Structured English, Decision Table or Decision Tree required. No unresolved issues have been identified in this process.

Table 14: Process Specification Process 5.3

Process Specification : Process 5.3	
Number	5.3
Name	Trigger System Alerts
Description	Receive routine schedule timings and M-CHAT-R follow-up data, evaluates notification conditions and delivers targeted notifications to the Parents.
Input Data Flow	Active Schedule Timings (from Process 5.1) M-CHAT-R Follow-Ups (from Process 5.2)
Output Data Flow	Routine Transition Reminders (to Parents) Follow-up Screening Alerts (to Parents)
Type of Process	☑ Online ☐ Batch ☐ Manual

Process Logic	IF Active Schedule Timings data flow is received THEN
	Construct specific routine transition message
	OUTPUT Routine Transition Reminder to Parents
	ENDIF
	
	IF M-CHAT-R Follow-ups data flow is received THEN
	Construct M-CHAT-R re-assessment follow-up alert message
	OUTPUT Follow-up Screening Alerts to Parents
	ENDIF

Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 14 presents the process specifications of Process 5.3. As an online process, it receives notification triggers from routine schedules and screening follow-up activities. The system then determines the appropriate notification type and delivers the corresponding alert to the Parents. Since the process involves generating and delivering notifications, no additional Structured English, Decision Table or Decision Tree required. There are no unresolved issues in this process.

Table 15: Process Specification Process 6.1

Process Specification : Process 6.1	
Number	6.1
Name	Compile Personal Tracking Data
Description	Gathers personal history profiles directly from the data store to format personal reports.
Input Data Flow	Child Profile (from D1: User & Profile Database)
Output Data Flow	Raw Personal Data (to Process 6.3)
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	FETCH child profile FROM D1: User & Profile Database Combine user information and historical records into a single dataset. Forward dataset as Raw Personal Data to Process 6.3
Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 15 shows the process logic of Process 6.1. The online process retrieves child profile information from User & Profile Database (D1). The system accesses the required profile records and prepares the information for further processing. Due to simple data retrieval procedure, no additional Structured English, Decision Table or Decision Tree required. No unresolved issues have been identified for this process.

Table 16: Process Specification Process 6.2

Process Specification : Process 6.2	
Number	6.2
Name	Compile User Data
Description	Combine routine progress records and screening logs to create research datasets.
Input Data Flow	Screening Result (from D2: Screening Database) Daily Routine Progress (from D3: Daily Routine Database)
Output Data Flow	Raw Anonymous Data (to Process 6.3)
Type of Process	☑ Online ☐ Batch ☐ Manual
Process Logic	FETCH screening result FROM D2 and daily routine progress FROM D3 Remove user identity details before processing the data Combine progress rates and screening scores for analysis Forward combined dataset as raw anonymous data to Process 6.3
Refer to Name	☐ Structured English ☐ Decision Table ☐ Decision Tree
Unresolved Issues	None

Table 16 describes the process specifications for Process 6.2. This online process collects screening data from database D2 and routine data from database D3. The system then combines the collected information and removes any personal details to create anonymous data for analysis purposes. Due to simple data collection, integration and anonymization procedures, no additional Structured English, Decision Table or Decision Tree required. There are no unresolved issues for this process

Table 17: Process Specification Process 6.3

<table>
<tr><th colspan="2">Process Specification : Process 6.3</th></tr>
<tr><td>Number</td><td>6.3</td></tr>
<tr><td>Name</td><td>Format Outputs & Dashboard</td></tr>
<tr><td>Description</td><td>Format raw datasets into a beautiful UI analytics dashboard, printable PDFs and stores formal diagnostic results into an analytics database.</td></tr>
<tr><td>Input Data Flow</td><td>Raw Personal Data (from Process 6.1)
Raw Anonymous Data (from Process 6.2)
Analytics & User Search Query (from NGO)</td></tr>
<tr><td>Output Data Flow</td><td>Printable PDF reports (to Parents)
Statistical Dashboard & User Summaries (to NGO)
Personal Report (to D4: Report & Analytics Database)</td></tr>
<tr><td>Type of Process</td><td>☑ Online
☐ Batch
☐ Manual</td></tr>
<tr><td>Process Logic</td><td>RECEIVE Raw Personal Data
RECEIVE Raw Anonymous Data
Combine raw personal data with raw anonymous data

Format the combined personal and statistical details
Generate printable PDF reports
Output printable PDF reports to parents

Extract the child progress history from the combined datasets
WRITE personal report to D4: Report & Analytics Database

IF Analytics & User Search Query is received from NGO THEN
    Filter the combined data based on the search options
    Compile the filtered statistic into charts and summaries
    OUTPUT Statistical dashboard & User Summaries to NGO interface
ENDIF</td></tr>
<tr><td>Refer to Name</td><td>☐ Structured English
☐ Decision Table
☐ Decision Tree</td></tr>
<tr><td>Unresolved Issues</td><td>None</td></tr>
</table>

Table 17 outlines the process logic of Process 6.3. This online process generates reports and analytics based on the personal and anonymous data received from previous processes. The system creates printable PDF reports for Parents, stores report records in database D4 and prepares statistical data for display on the NGO dashboard. It also supports data filtering based on search criteria to provide relevant analytical information. Since the process follows a simple of the report generation and analytics procedures, no additional Structured English, Decision Table or Decision Tree required. No unresolved issues have been identified for this process.

7.0 Physical System Design

7.1 Physical DFD TO-BE System

7.1.1 Diagram 0

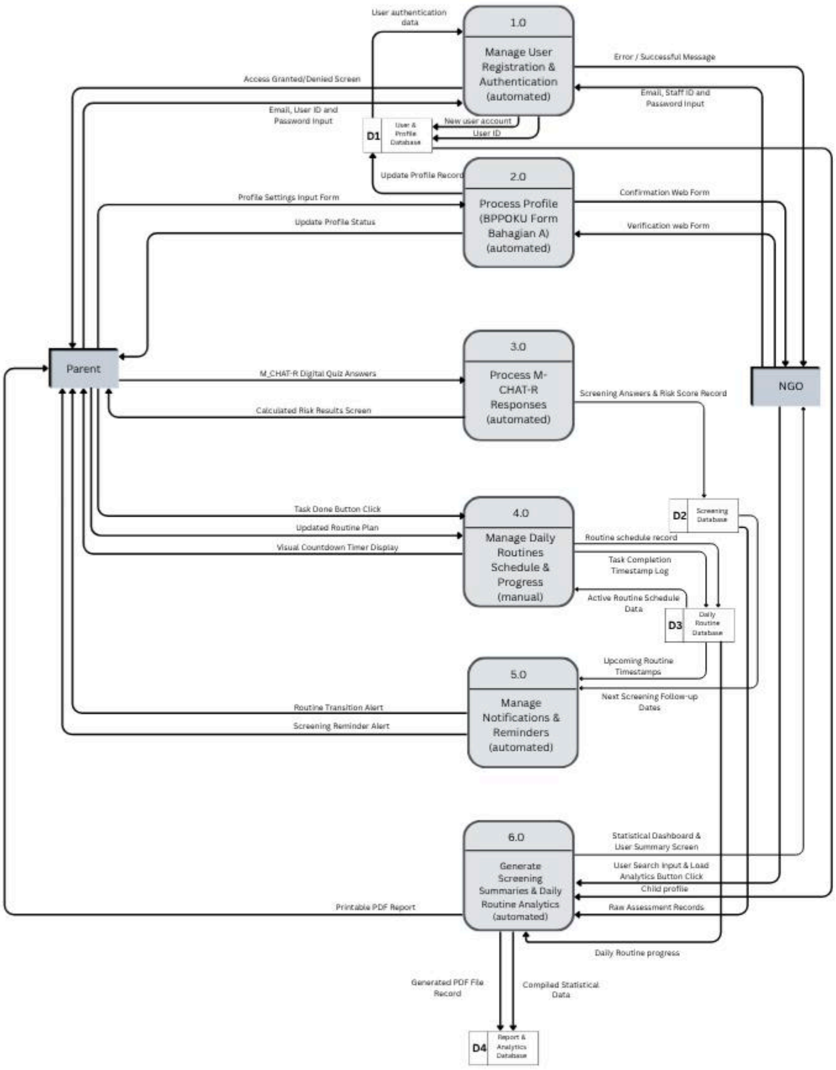

Figure 12: Figure shows the broad architectural boundaries of the system

This figure visually groups the system's core functions into distinct technological modules, explicitly indicating which processes handle sensitive user authentication, execute automated risk calculations, render active visual screens, run automatically as background servers, and manage heavy database reporting.

Link for clearer picture: https://canva.link/s5jcmrlcm81sz2x

7.1.2 Child Diagram

- Manage User Registration & Authentication

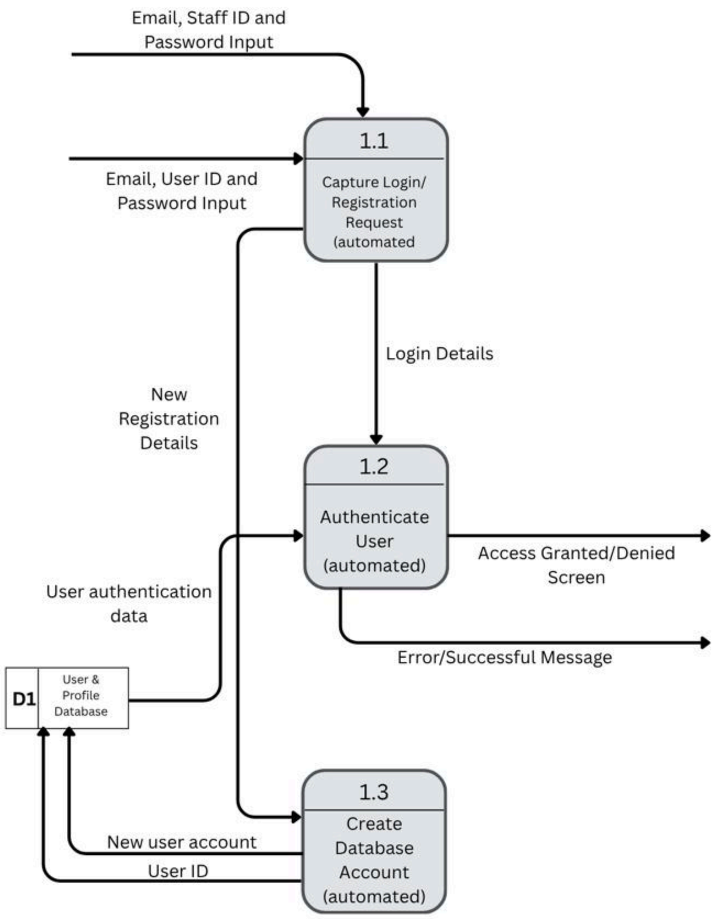

Figure 13: Figure shows the child diagram of manage user registration and administration (1.0)

This figure breaks down the system's login and security implementation. It shows the front-end interface capturing the user's email and password inputs, which are then passed to a secure back-end authentication script. The script queries the User & Profile Database (D1) to verify credentials and outputs either an access granted screen or an error message.

- Process Profile

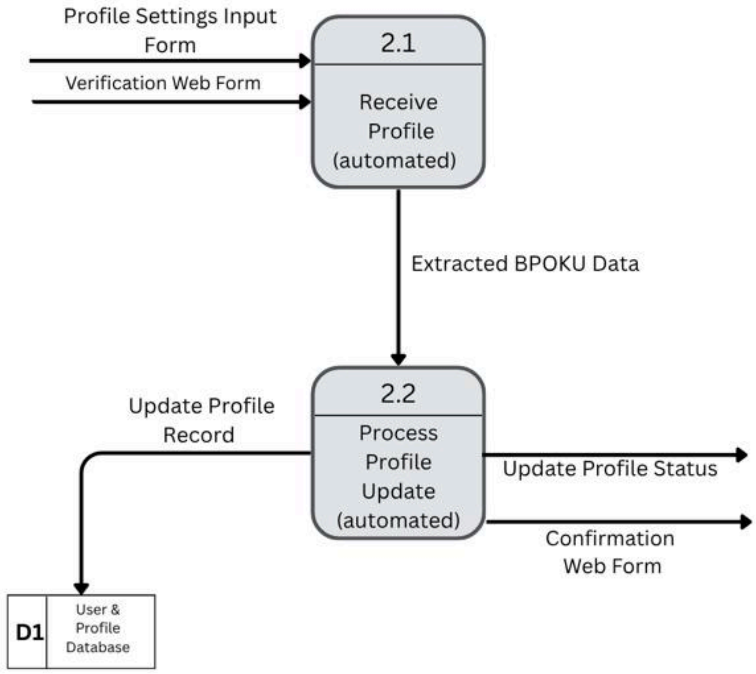

Figure 14: Figure shows the child diagram of process profile (2.0)

This figure explains the physical processing of BPPOKU data. It focuses on how the system extracts raw data from the profile settings input form, executes a database update to save the record in D1, and ultimately renders a confirmation web form to the NGO for verification.

- Process M-CHAT-R Responses

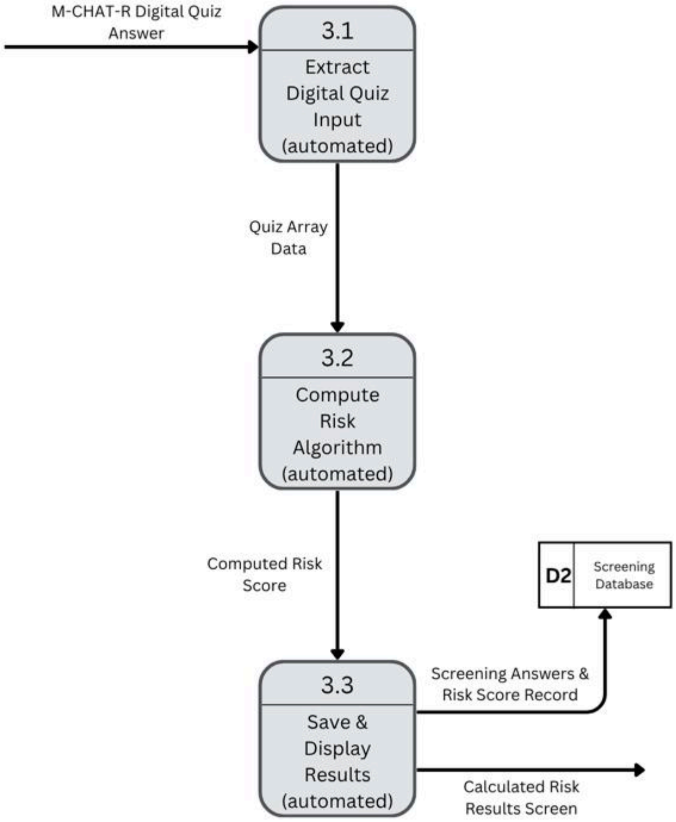

Figure 15: Figure shows the child diagram of process M-CHAT-R responses (3.0)

This figure details the digital autism screening tool. It shows the system capturing the parent's digital quiz answers, running a programmed math algorithm to compute the exact risk score, and then simultaneously saving the record to the Screening Database (D2) while rendering the calculated results screen.

- Manage Daily Routines Schedule & Progress

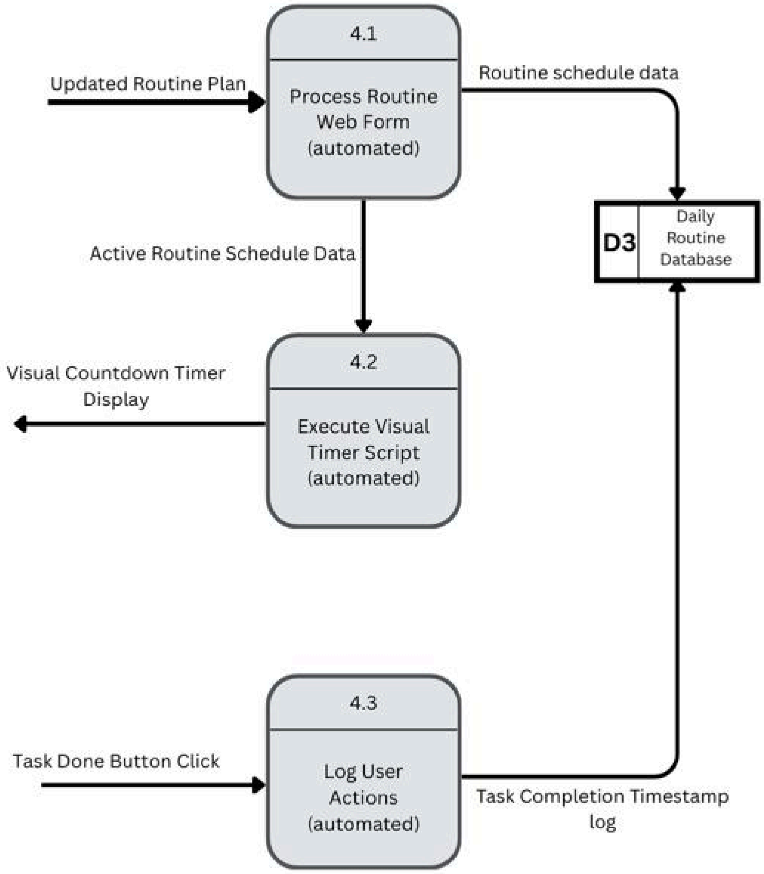

Figure 16: Figure shows the child diagram of manage daily routines schedule and progress (4.0)

This figure illustrates the physical execution of the routine tracker process. It shows how an updated routine plan is processed by the automated routine processing module (4.1) and stored as routine schedule data in the Daily Routine Database (D3). The active routine schedule data is then passed to the visual timer script (4.2), which generates the visual countdown timer display. Finally, when the user clicks the "Task Done" button, the automated logging process (4.3) records the task completion timestamp in the Daily Routine Database (D3).

- Manage Notifications & Reminders

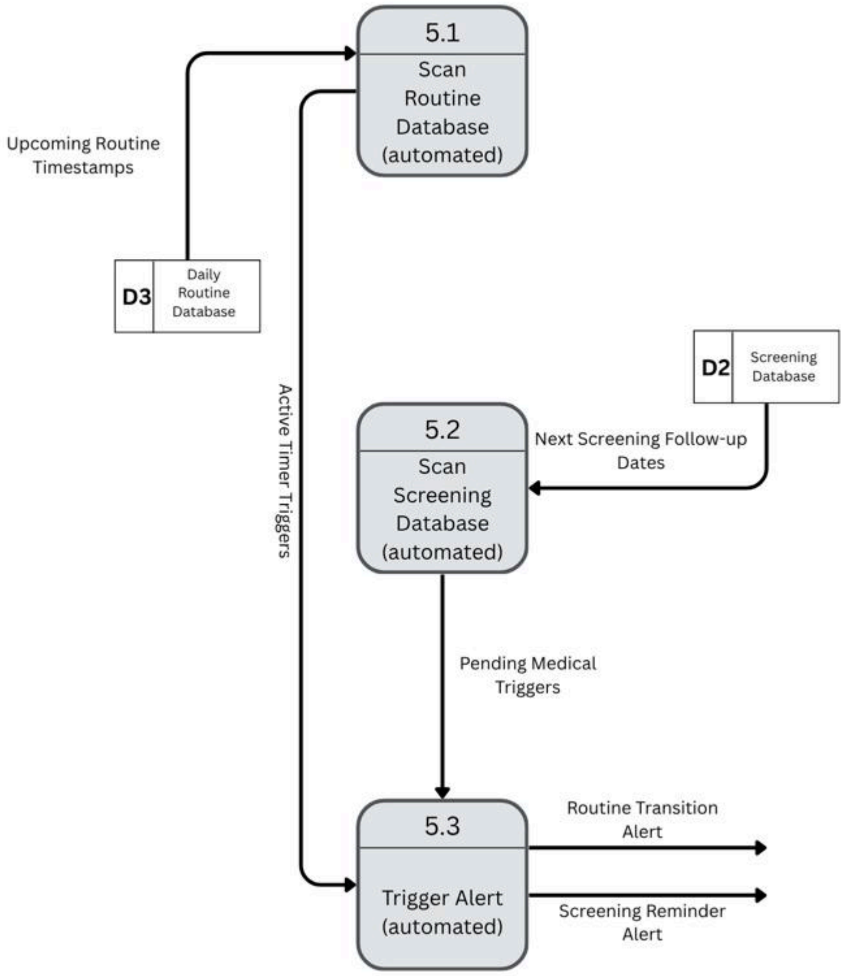

Figure 17: Figure shows the child diagram of manage notifications and reminders (5.0)

This figure shows the system's automated background services. It explains how server-side automatically scans the routine and screening databases for matching time conditions. When a specific timing is reached, the system executes an automated trigger to push routine transition or screening reminder alerts directly to the user's device.

- Generate Screening Summaries & Daily Routine Analytics

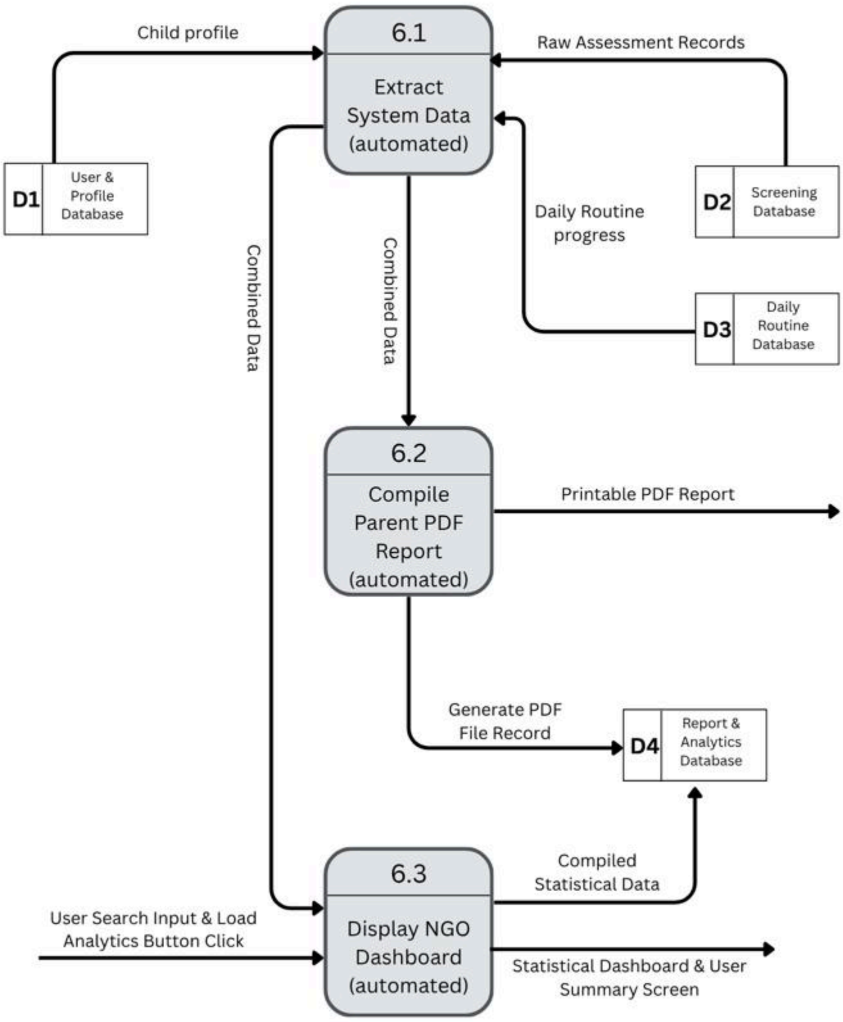

Figure 18: Figure show the child diagram of generate screening summaries and daily routine analytics (6.0)

This figure shows the details of the system's data extraction and rendering engines. It shows the shared utility process pulling raw data across databases to serve two distinct functions: executing a script to compile and generate a printable PDF file for the parent, and processing a user search query to load a dynamic, visual statistical dashboard screen for the NGO.

7.1.3 Partitioning

- Diagram 0

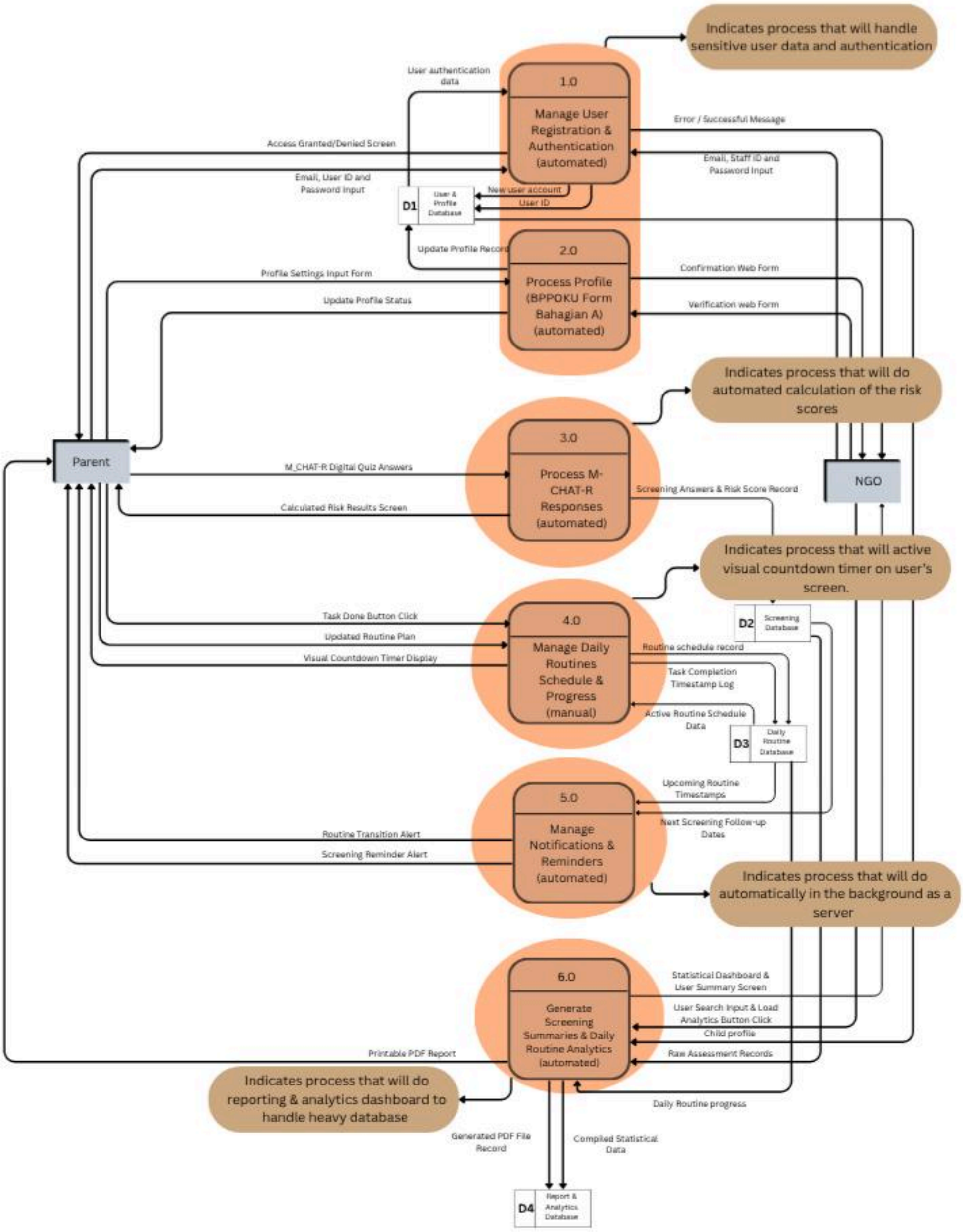

Figure 19: Figure shows the broad architectural boundaries of the system

This figure highlights the broad architectural boundaries of the system. The partitions visually group the system's core functions into distinct technological modules, explicitly indicating which processes handle sensitive user authentication, execute automated risk calculations, render active visual screens, run automatically as background servers, and manage heavy database reporting.

Link for clearer picture: https://canva.link/s5jcmrlcm81sz2x

- Child Diagram
- Manage User Registration & Authentication

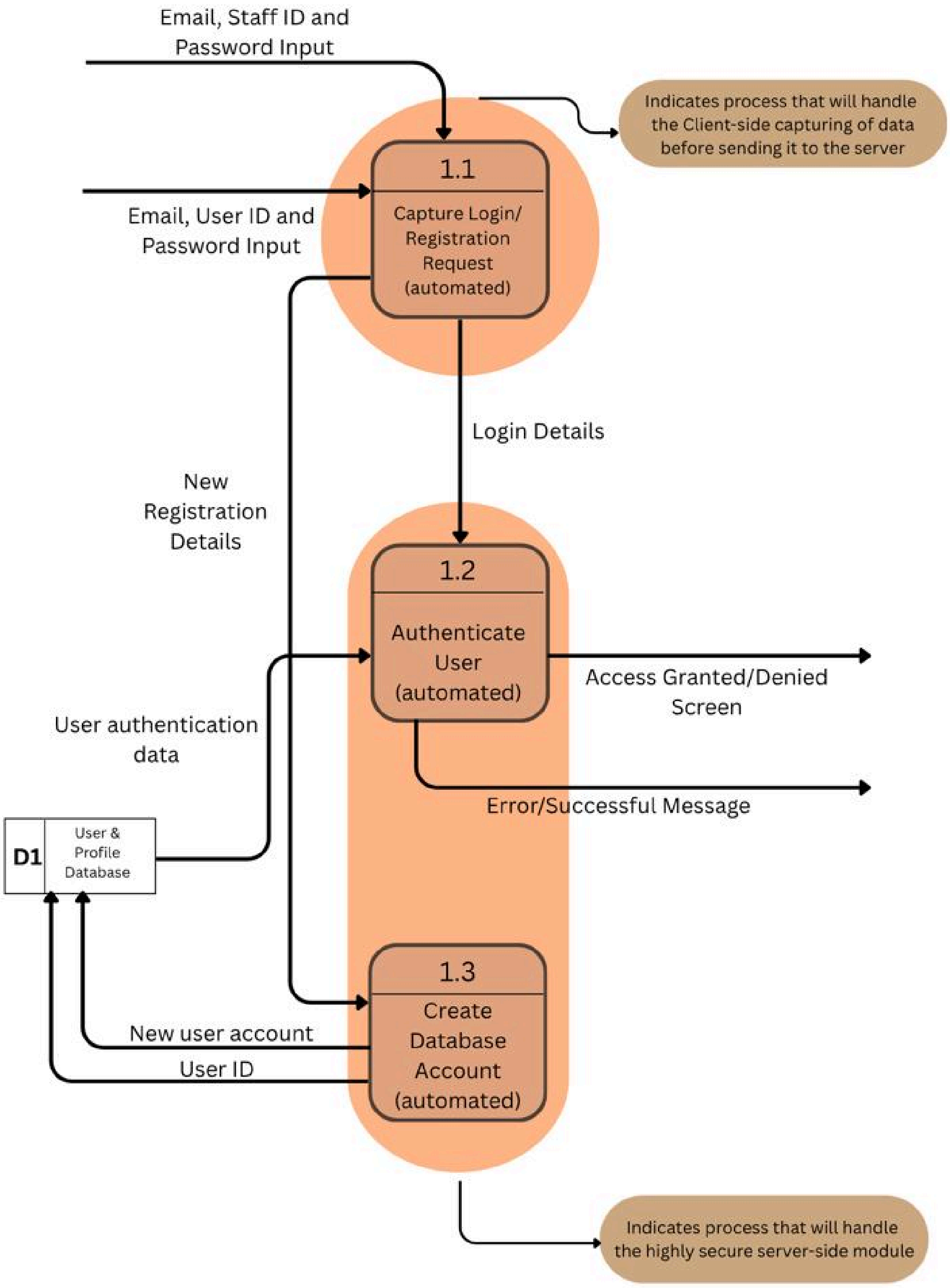

Figure 20: Figure shows the partitioning child diagram of manage user registration and administration (1.0)

This figure utilizes a strict client-server partition. Process 1.1 is partitioned as the client-side module responsible for capturing user data before sending it over the network. In contrast, Processes 1.2 and 1.3 are tightly grouped into a highly secure server-side module to protect sensitive authentication logic and database creation.

- Process Profile

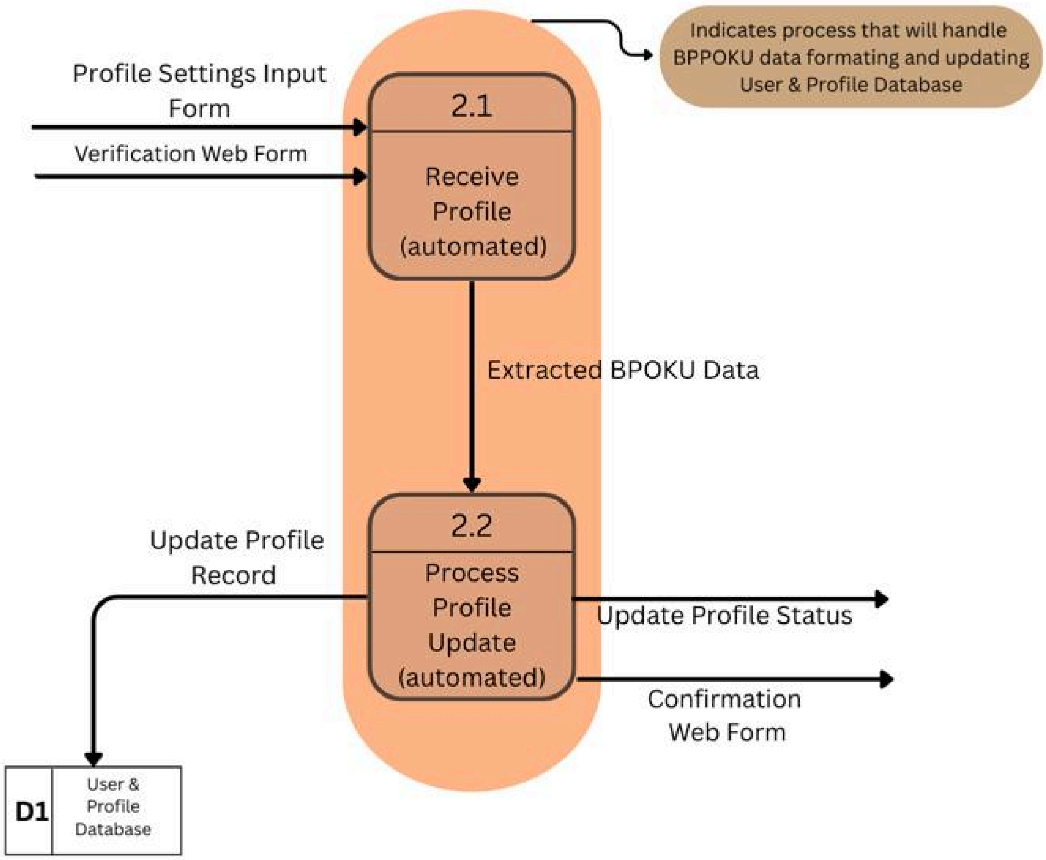

Figure 21: Figure shows the partitioning child diagram of process profile (2.0)

This figure shows that processes 2.1 and 2.2 are encapsulated into a single unified partition. This indicates a dedicated back-end module specifically engineered to handle BPPOKU data formatting and safely execute updates to the User & Profile Database.

- Process M-CHAT-R Responses

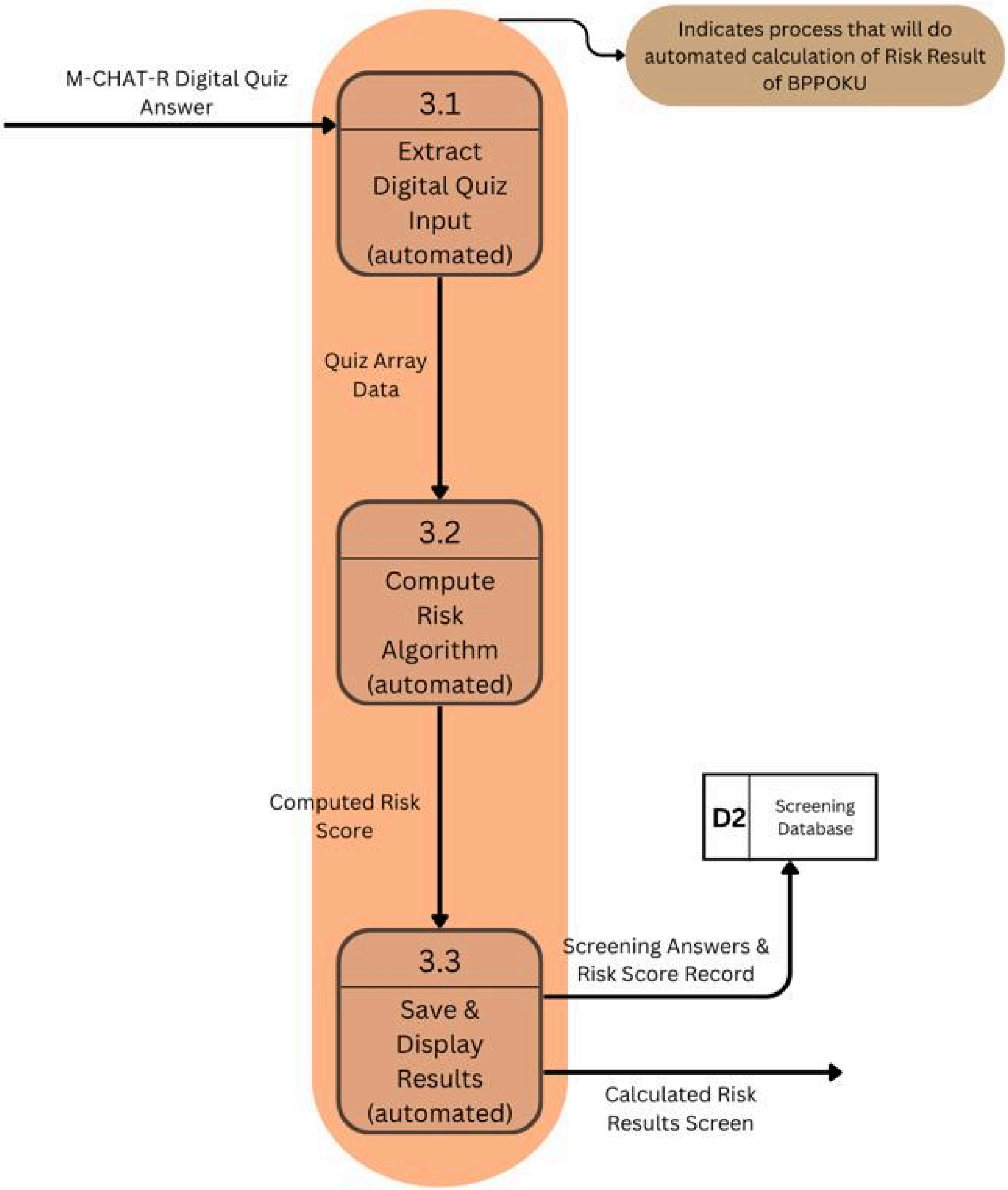

Figure 22: Figure shows the partitioning child diagram of process M-CHAT-R responses (3.0)

This entire child diagram is grouped into one continuous partition. This indicates that extracting digital quiz input, computing the algorithm, and saving the results are all executed together as a single, uninterrupted automated calculation module to generate the Risk Result of BPPOKU.

- Manage Daily Routine Schedule & Progress

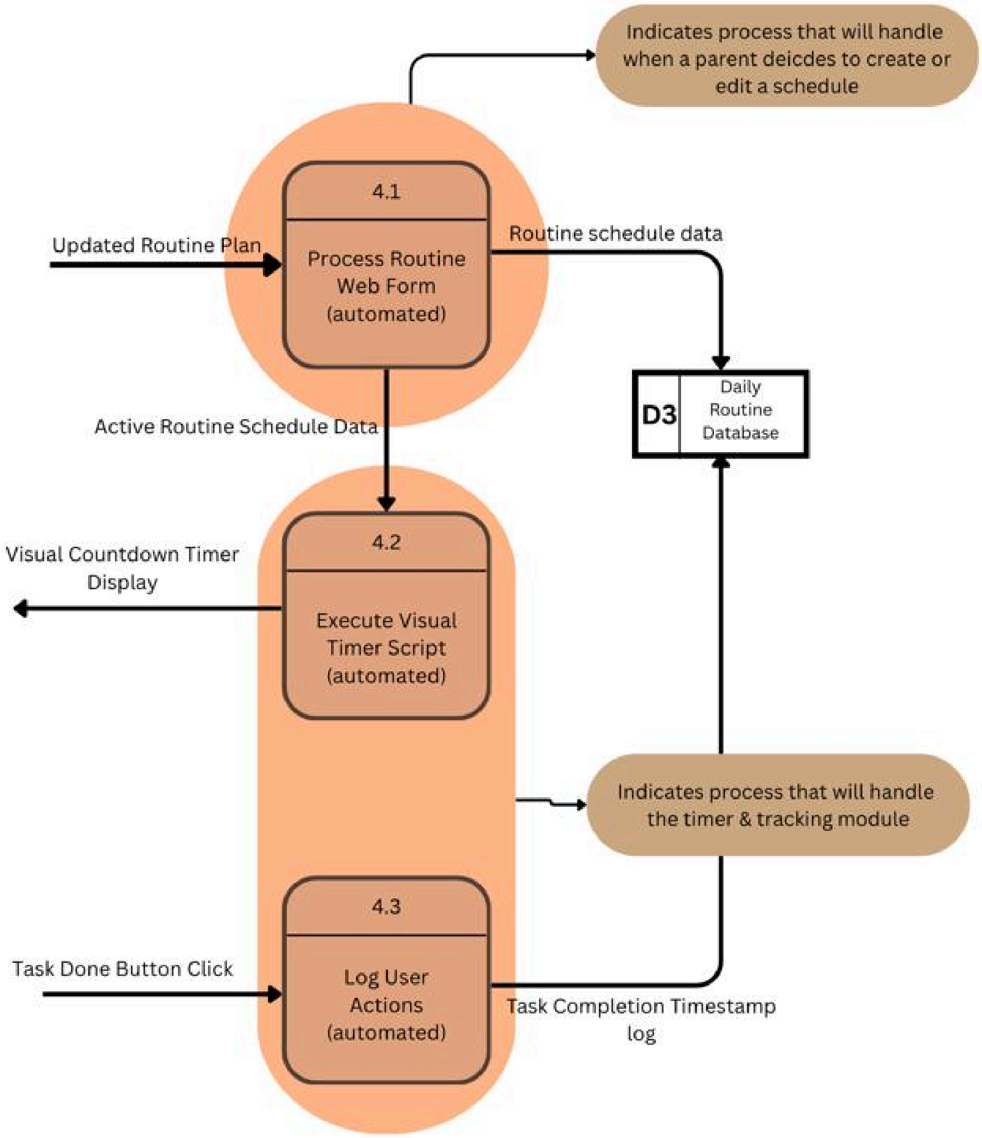

Figure 23: Figure shows the partitioning child diagram of manage daily routines schedule and progress (4.0)

This figure shows that process 4.1 is isolated to indicate the module that handles when a parent decides to create or edit a schedule. Separately, Processes 4.2 and 4.3 are grouped together to represent the active timer and tracking module that runs continuously on the user's screen.

- Manage Notifications & Reminders

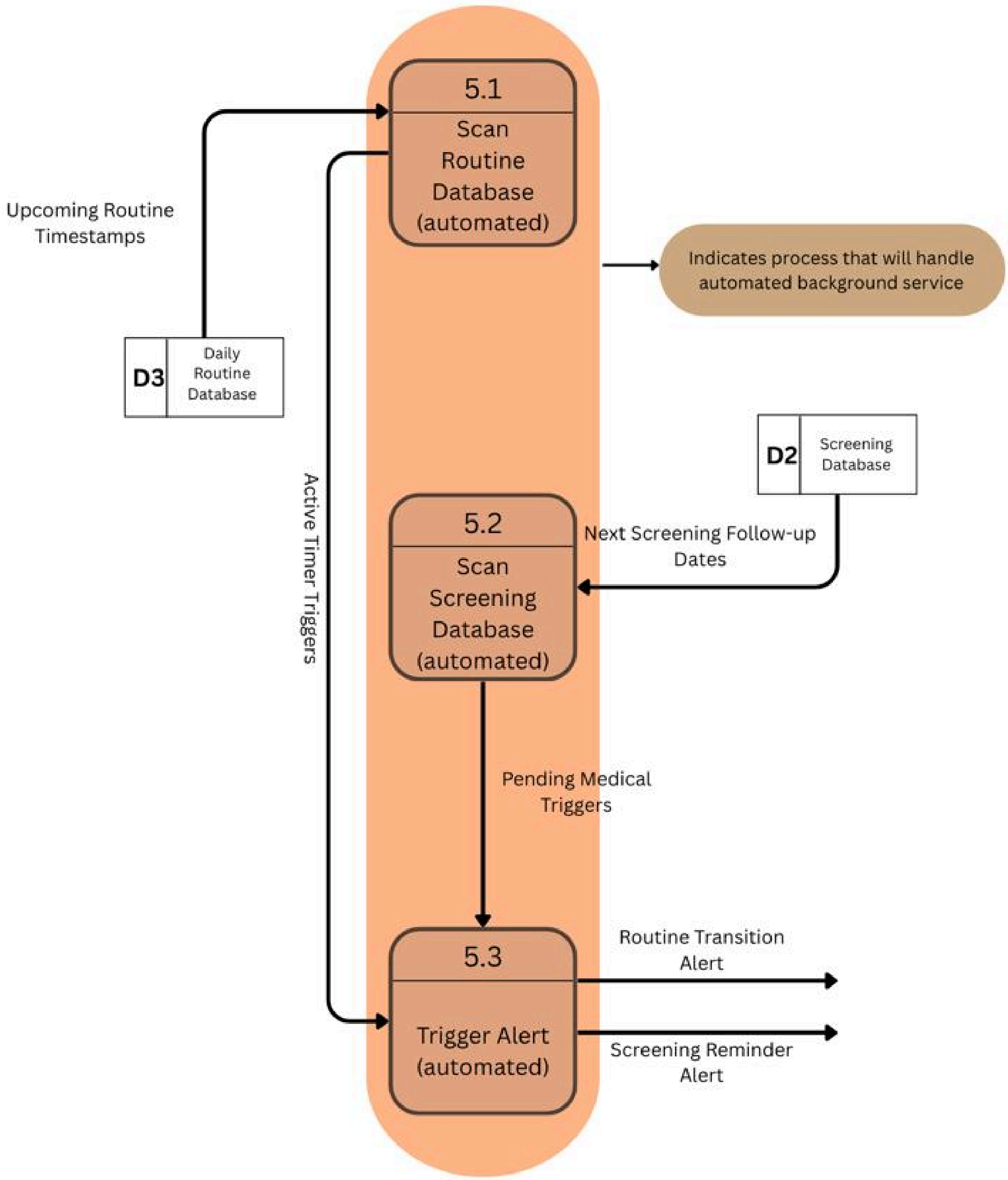

Figure 24: Figure shows the partitioning child diagram of manage notifications and reminders (5.0)

Processes 5.1, 5.2, and 5.3 are entirely enclosed within a single partition. This illustrates that these database scans and alert triggers function together as an automated background service, operating entirely on the server without any manual user interaction.

- Generate Screening Summaries & Daily Routine Analytics

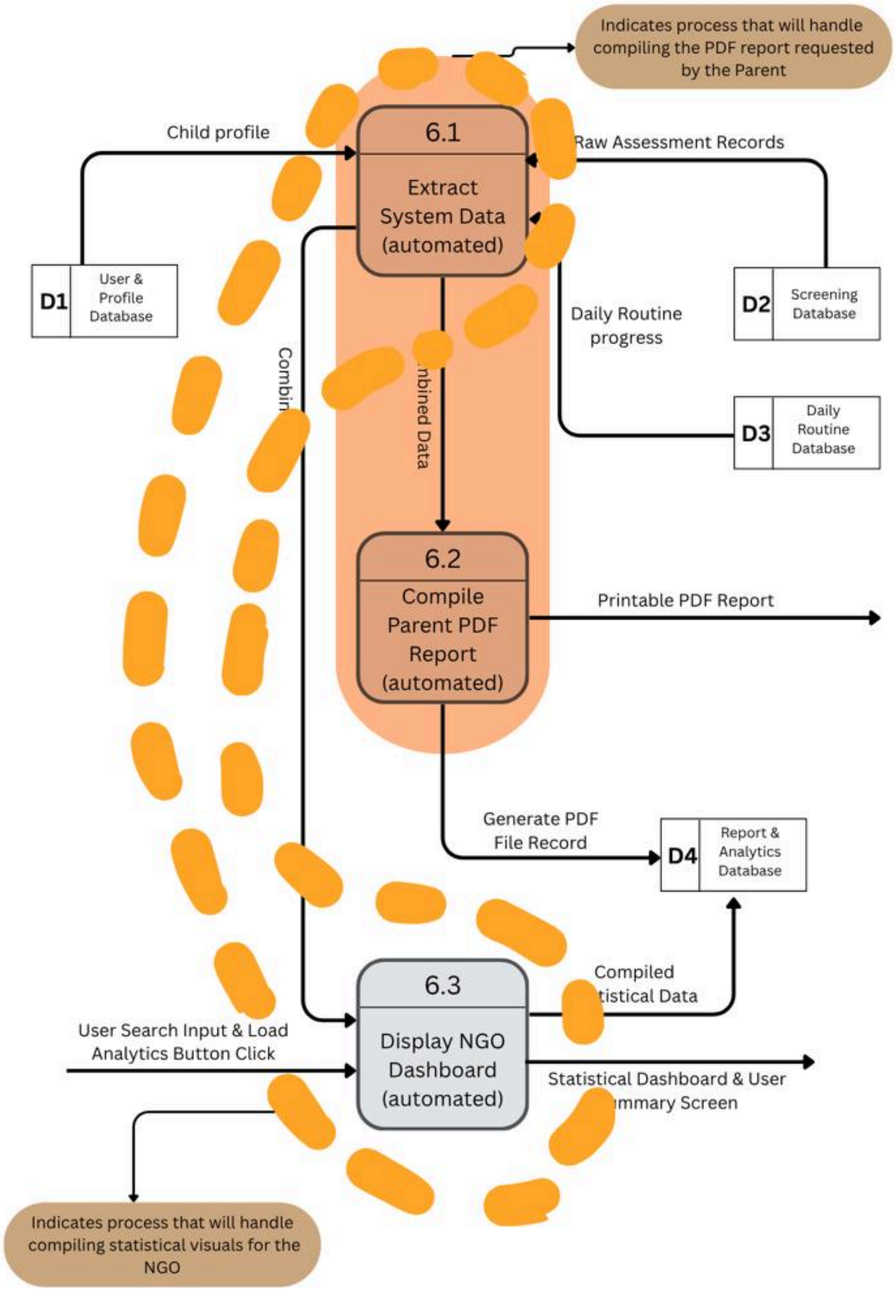

Figure 25: Figure show the partitioning child diagram of generate screening summaries and daily routine analytics (6.0)

This figure utilizes an advanced overlapping partition strategy to demonstrate shared system resources. The solid orange partition illustrates the specific execution path for compiling the PDF report requested by the Parent, linking data extraction (6.1) directly to PDF generation (6.2). Simultaneously, the dotted orange partition reveals that the same data extraction process (6.1) is also utilized by a separate execution path to handle compiling statistical visuals for the NGO (6.3).

7.1.4 CRUD Matrix

Activity	User & Profile DB (D1)	Screening DB (D2)	Daily Routine DB (D3)	Report & Analytics DB (D4)
User Logon	R	-	-	-
Register New Account	C	-	-	-
Update & Verify Profile	RU	-	-	-
Delete User Account	D	-	-	-
Submit M-CHAT-R Assessment	-	C	-	-
View Assessment Results	-	R	-	-
Setup Routine Schedule	-	-	C	-
Track / Update Task Completion	-	-	RU	-
Remove Old Routine Schedule	-	-	D	-
Trigger System Notifications	-	R	R	-
Generate Personal PDF Report	R	R	R	C
View NGO Analytics Dashboard	R	R	R	R

Table 18: Table shows the CRUD matrix of the system

The CRUD Matrix provides a strict blueprint of how every major system activity interacts with our backend data storage. It maps out exactly which system functions Create (C), Read (R), Update (U), or Delete (D) data across our four core databases: the User & Profile DB (D1), Screening DB (D2), Daily Routine DB (D3), and Report & Analytics DB (D4). For instance, when a parent sets up a new routine schedule, the system executes a "Create" operation exclusively in D3, but when a parent requests a personal PDF report, the system must "Read" data from D1, D2, and D3 before "Creating" the final file record in D4. Notably, when the NGO views the analytics dashboard, the system reads live data across all four databases to compile accurate user summaries. This matrix proves that our database relationships are fully mapped and logically sound.

7.1.5 Event Response Table

Event	Source	Trigger	Activity	Response	Destination
User logs on or registers	Parent / NGO	Email and Password Input / User ID and Password	Verify credentials. Create a new user account in the database if new. Generate user ID.	Access Granted / Denied Screen, Error / Successful Message	Parent / NGO
Parent updates profile	Parent	Profile Settings Input Form	Update user profile record in the database.	Update Profile Status	Parent
NGO verifies user profile	NGO	Verification Web Form	Verify parent profile data and update status in the database.	Confirmation Web Form	NGO
Parent submits M-CHAT-R quiz	Parent	M-CHAT-R Digital Quiz Answers	Compute automated risk score. Store screening answers and risk score record.	Calculated Risk Results Screen	Parent
Parent sets up or completes a routine	Parent	Updated Routine Plan / Task Done Button Click	Store a new routine schedule. Update task completion timestamp log. Operate visual timer.	Visual Countdown Timer Display	Parent
Scheduled routine or medical date is reached	Temporal (System Time)	Upcoming Routine Timestamps / Next Screening Follow-up Dates	Scan routine database and screening database for matching times. Trigger mobile alerts.	Routine Transition Alert, Screening Reminder Alert	Parent
Parent requests personal report	Parent	Report request	Retrieve child profile, raw assessment records, and daily routine progress. Compile PDF.	Printable PDF Report	Parent

Event	Source	Trigger	Activity	Response	Destination
NGO views system analytics	NGO	Load Analytics / User Search Query	Retrieve aggregated anonymous data. Compile statistical data and generate charts.	Statistical Dashboard & User Summaries	NGO

Table 19: Table shows the event response table

The Event Response Table acts as the master operational rulebook for the system. It documents every single external and temporal event that interacts with our application, ensuring there are no dead ends in the user experience. For each event, it details the Source (such as a Parent, NGO, or System Time) and the physical Trigger (like an email/password input, a button click, or an upcoming timestamp). It then defines the exact Activity the system must perform,such as verifying credentials or computing an automated risk score and the final Response delivered back to the Destination. Whether it is the system triggering a mobile alert for an upcoming routine, or an NGO querying the database to receive a Statistical Dashboard & User Summaries, this table guarantees that every trigger has a clearly defined, automated system reaction.

7.1.6 Structure Chart

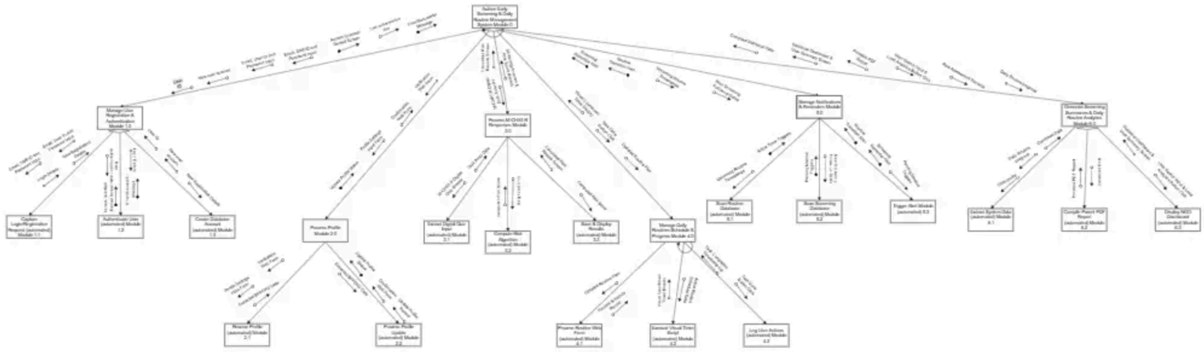

Figure 26: Figure shows the structure chart of the system

The Structure Chart translates our system's logical and physical data flows into a concrete, hierarchical software architecture. At the top, the central system module acts as the main controller, branching down into the six primary executable program modules (such as the Manage User Registration module, the Process M-CHAT-R module, and the Generate Reports module). These primary modules further break down into specific, single-function sub-modules. The arrows between these boxes illustrate the exact execution paths, showing how data couples (parameters) and control flags (status conditions) are passed back and forth between parent and child modules. Essentially, this diagram is the direct blueprint for the development team, detailing exactly how the backend scripts and program functions need to be coded and organized.

Link for clearer picture:
https://drive.google.com/file/d/1Dxk3IqCmsgbDo2XLgRZMFVITVhIMyiX0/view?usp=sharing

7.1.7 System Architecture

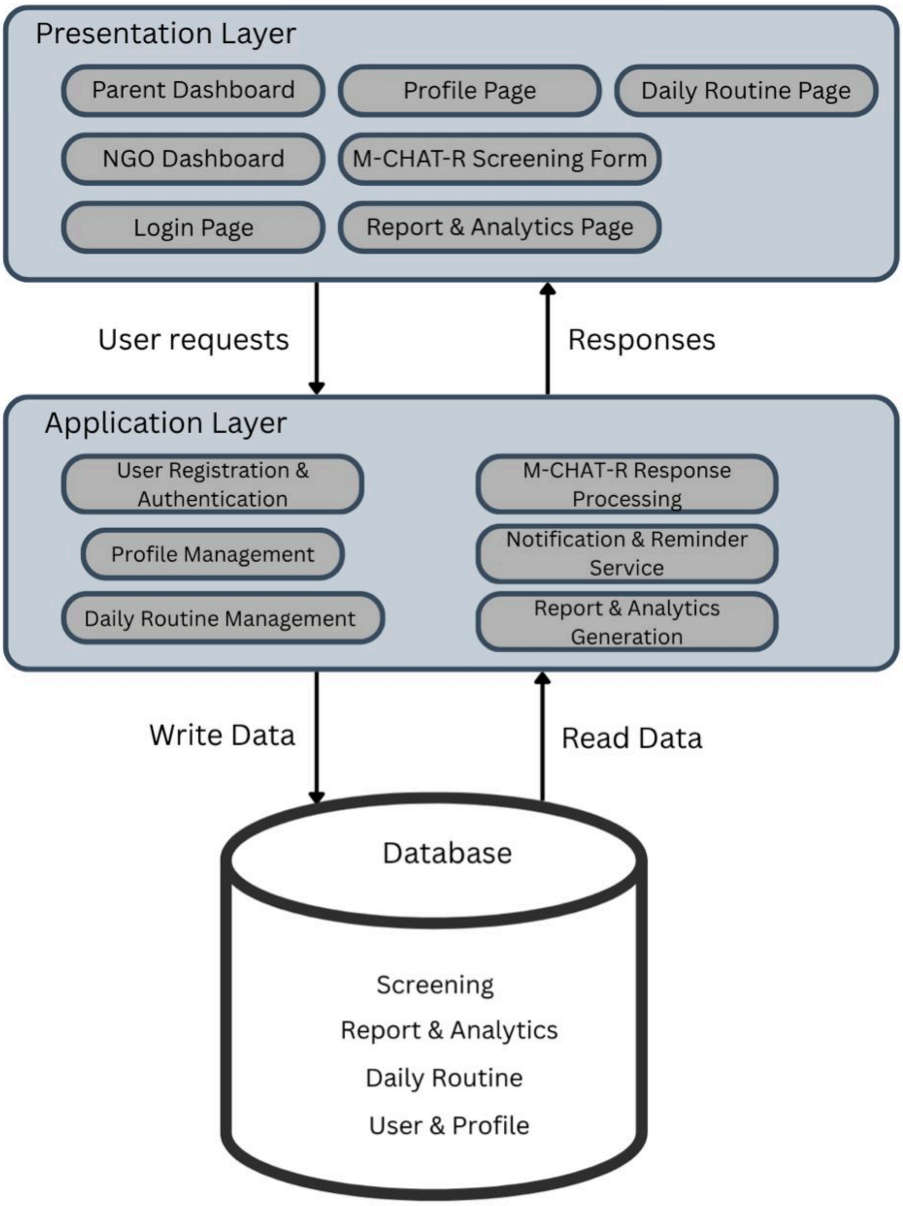

Figure 27: Figure shows the system architecture

The architecture follows a three-tier structure comprising the Presentation Layer (user interfaces for parents and NGO), the Application Layer (core system logic including M-CHAT-R processing, authentication, notification services, profile management, daily routine management and report generated), and the Database Layer (persistent storage for screening results, analytics data, daily routines, and user profile). User requests flow from the Presentation Layer to the Application Layer, which reads from and writes to the underlying database to generate appropriate responses.

8.0 System Wireframe (Input Design, Output Design)

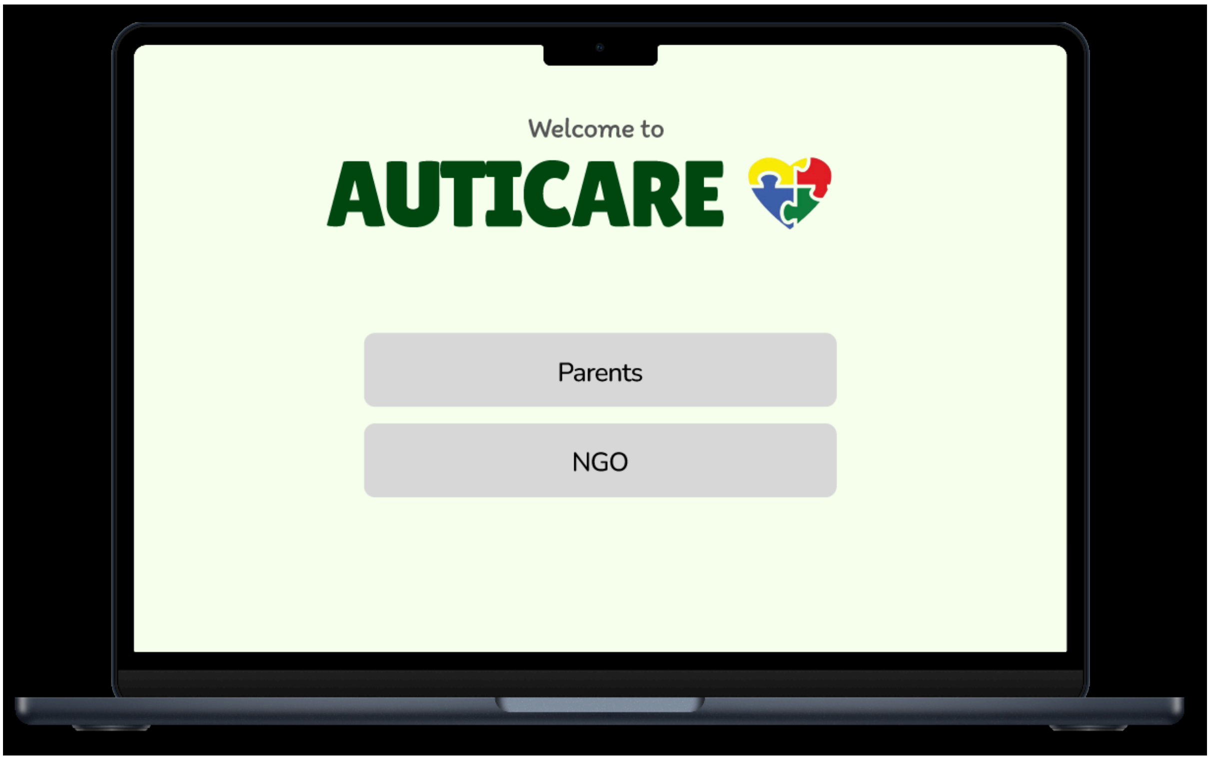

Figure 28: Figure shows the main screen for both entities

This figure displays the interface that allows users to select their account role as either a Parent or an NGO. Each option directs users to different sections of the system based on their responsibilities and access permissions, as both entities have different functions and roles within the system.

- NGO

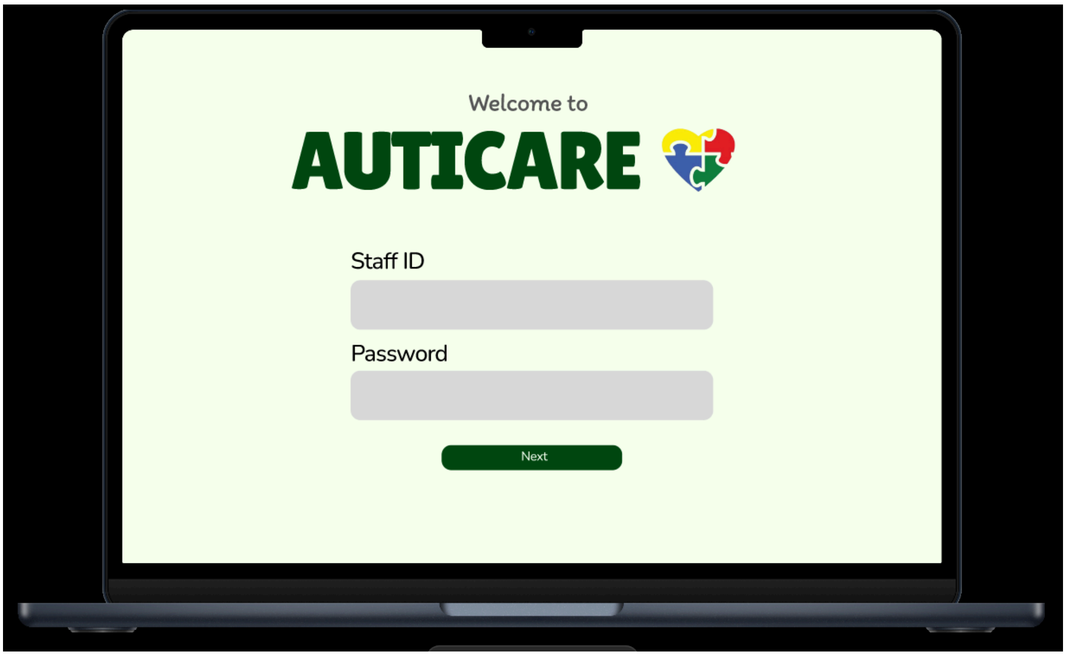

Figure 29: Figure shows the main screen for NGO

This figure displays the login interface that appears when the user selects an NGO. The system prompts the user to enter the required login credentials to verify their identity before granting access to the NGO-related features and services.

Figure 30: Figure shows the menu for NGO

This figure displays the navigation menu provided for NGO users, allowing them to select the required task or function they want to perform. Once a selection is made, the system will directly proceed to the corresponding operation, enabling NGOs to manage user records, messages, screening summaries, and other system activities efficiently.

Figure 31: Figure shows reports and analytics for all user

This figure displays the overview of screening results and weekly routine progress records for all registered users. The feature is designed to support multiple children under one parent account, allowing parents with more than one child to view and manage each child's screening summary and routine progress separately. This enables the system to provide a more flexible and organized way to monitor the development and activities of each child.

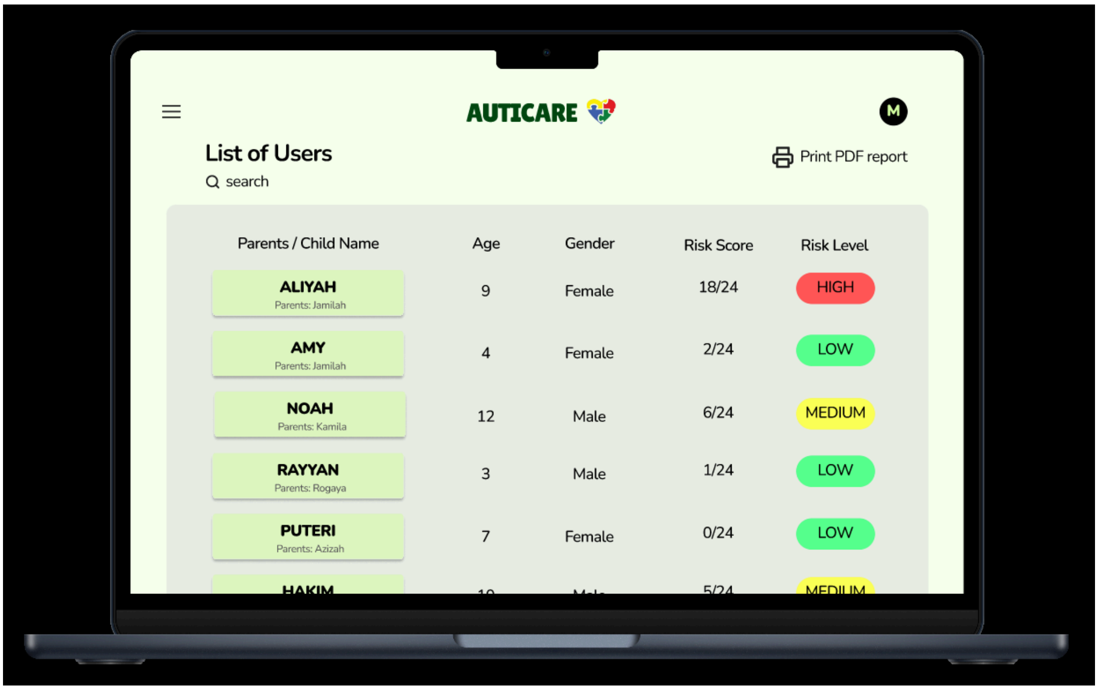

Figure 32: Figure shows the list of users with their details

This figure displays the user information records managed by the system, including details about parents and their children. The information provided includes the child's demographic details, M-CHAT-R risk score, and risk level, allowing the NGO to monitor screening results and review user records efficiently.

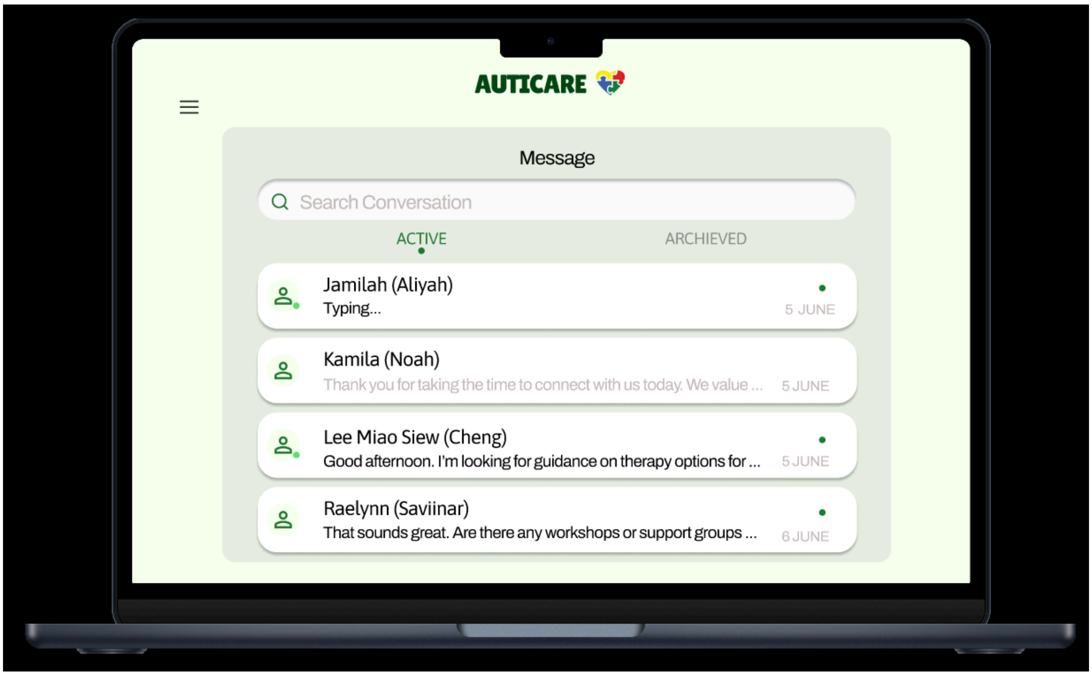

Figure 33: Figure shows list of message from the chatbox

This figure displays the inbox interface of the NGO, showing messages sent by users for communication and support purposes. The system indicates messages that have already been read by displaying them in a faded appearance, allowing the NGO to easily differentiate between unread and previously viewed messages.

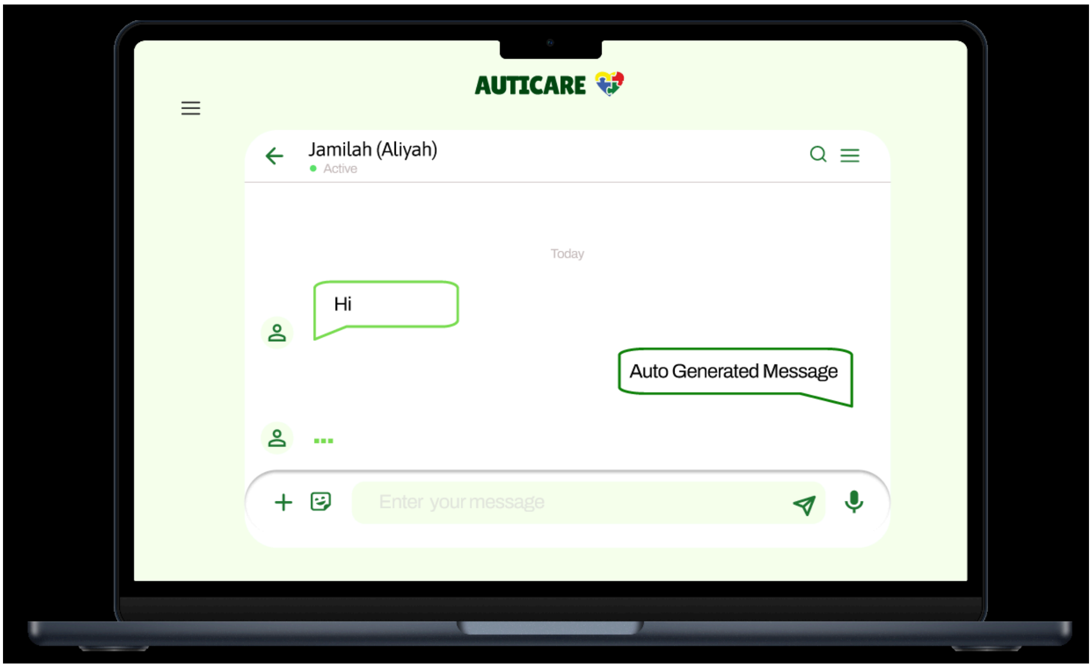

Figure 34: Figure shows message between Jamilah and NGO

This figure shows a live message screen between the Parents, Jamilah and her daughter Aliyah and the NGO admin. This feature allows users to ask any questions to the admin at any time as it is a 24-hour service.

- Parents

Figure 35: Figure shows the main screen for parents

This figure displays the authentication interface where existing users are prompted to enter their user ID and password to access the system. For new users, the system provides an option to create a new account, allowing them to complete the registration process before using the system features.

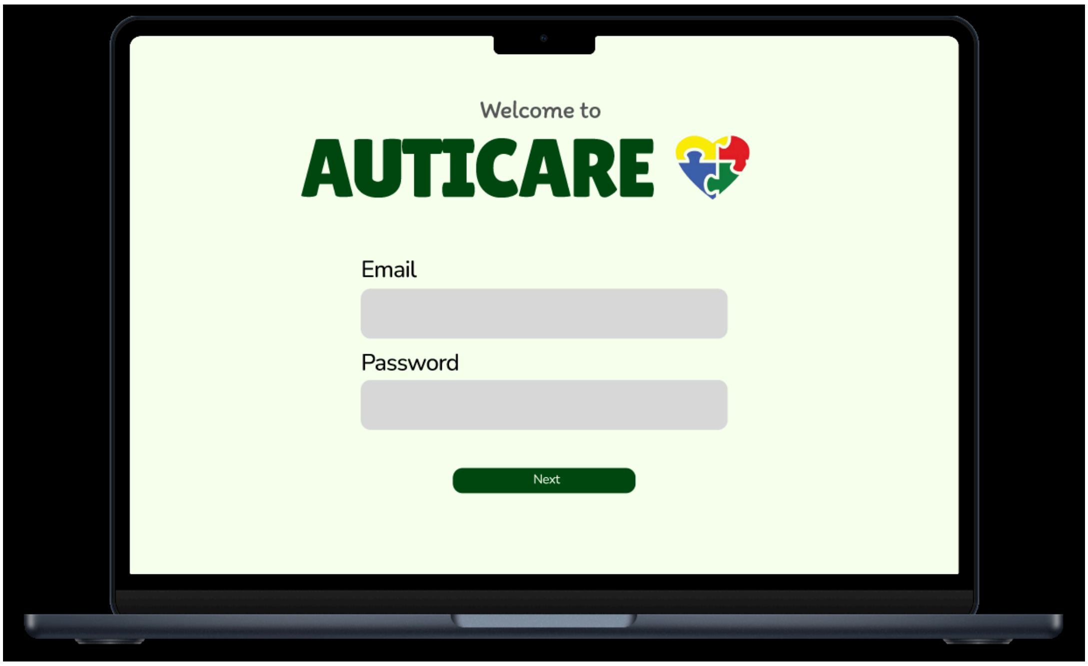

Figure 36: Figure shows the main screen for parents

This figure displays the registration interface when the user selects the option to create a new account. The system prompts users to provide their email address and password as the required information to proceed with the account registration process.

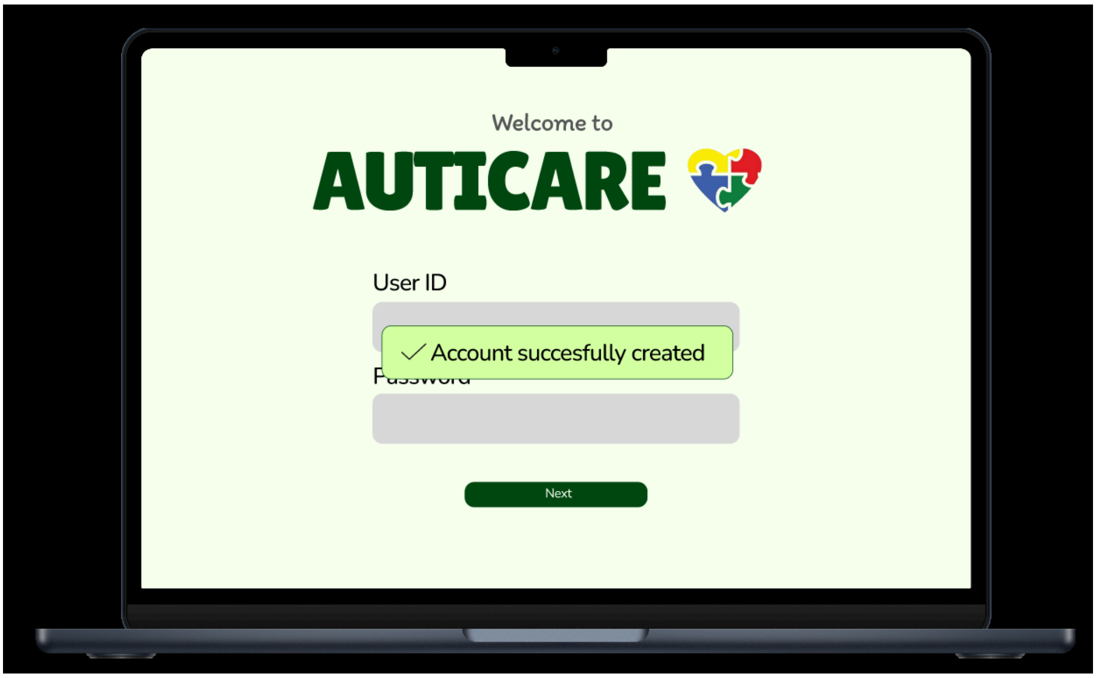

Figure 37: Figure shows the successful message

This figure displays the confirmation message after the user completes the account registration process. The system informs the user that the account creation is successful and provides the user ID through email for future login and account access.

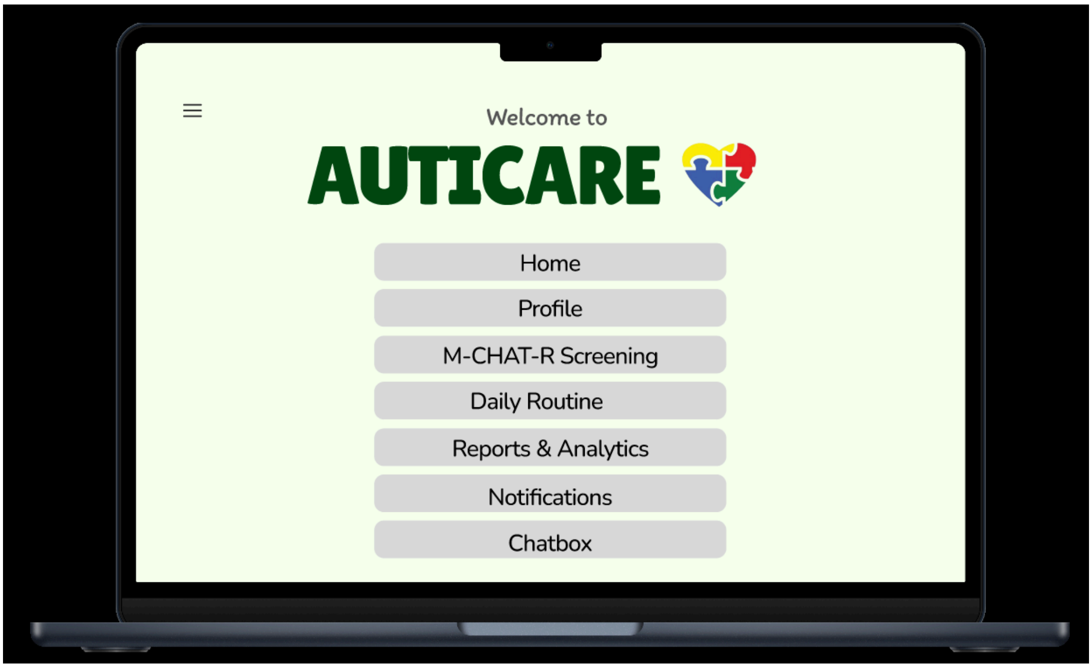

Figure 38: Figure shows the menu for parents

This figure displays the navigation menu provided for parents to access different features within the system. Users can select various options such as Home, Profile, M-CHAT-R Screening, Daily Routine, Reports and Analytics, Notifications, and Chatbox. The menu allows parents to easily navigate through the system and access the required functions efficiently.

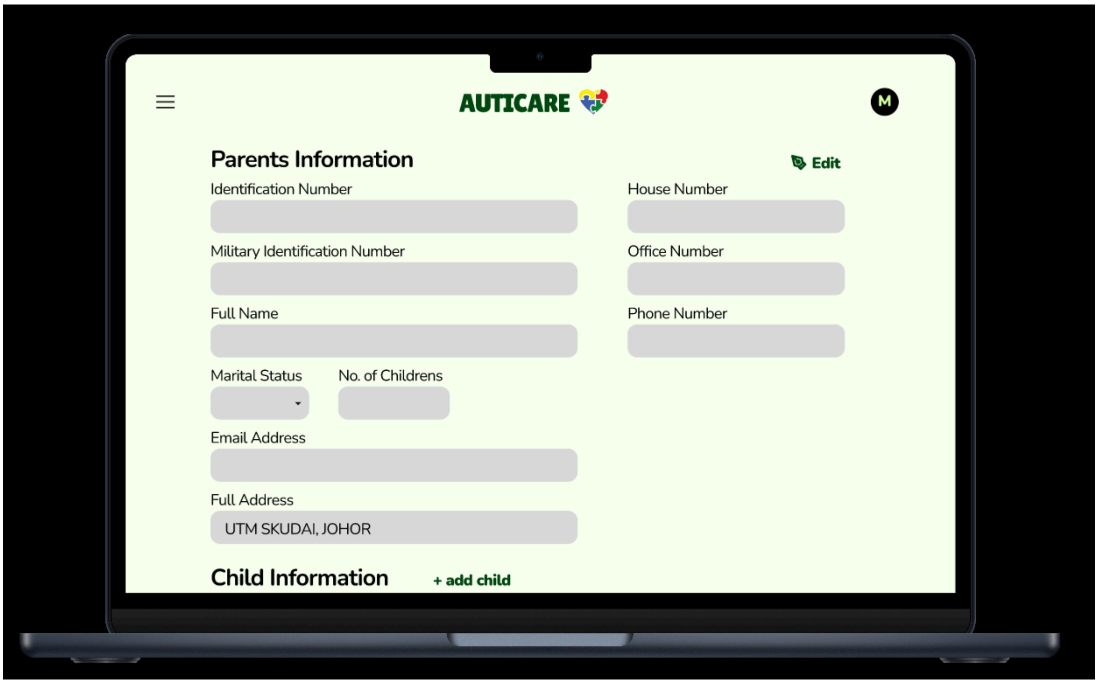

Figure 39: Figure shows the profile screen

This figure displays the BPPOKU registration form where parents are required to provide their personal details and their child's information. The form allows the system to collect the necessary data for registration, identification, and further assessment processes.

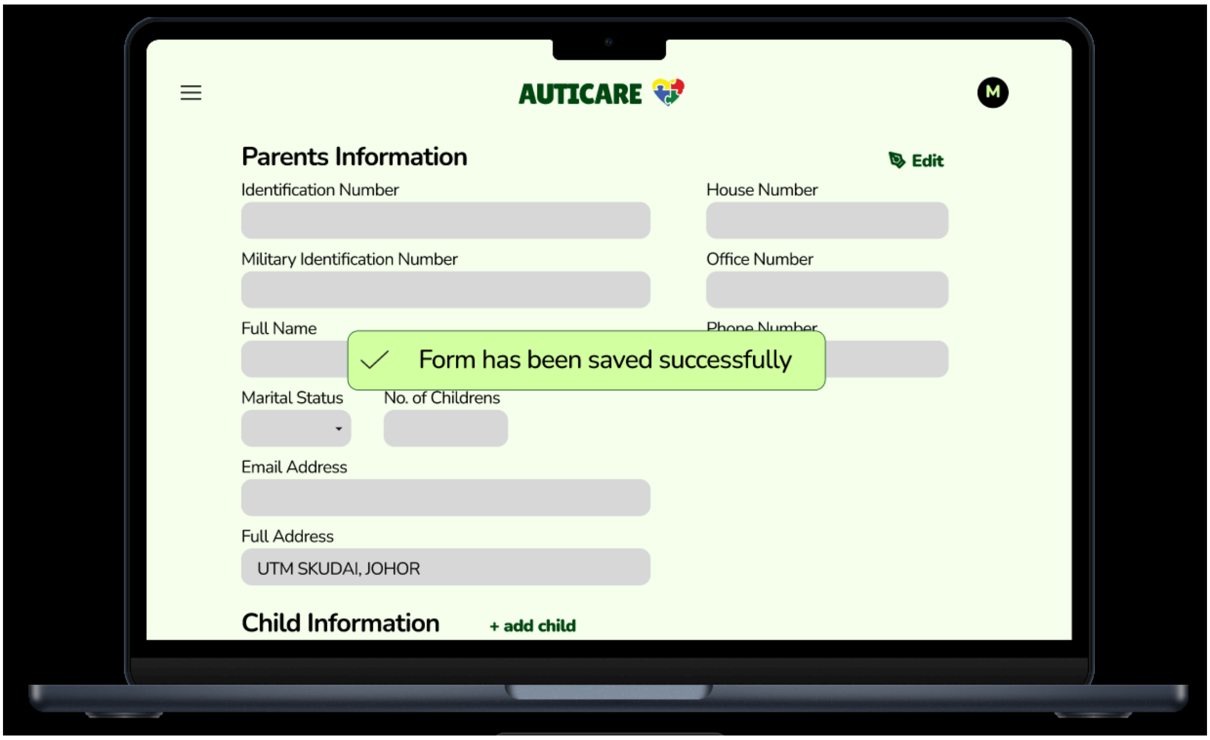

Figure 40: Figure shows the successful message

The figure illustrates the successful message displayed after the user's action has been completed. It confirms that the system has processed the request successfully and provides feedback to the user that the operation has been completed without any errors.

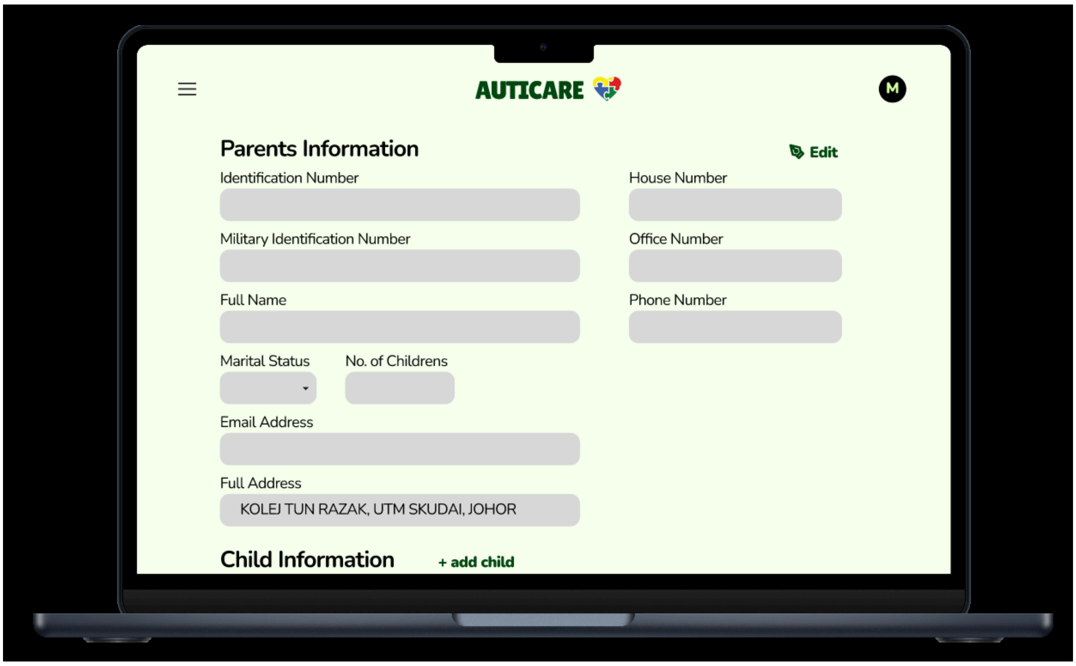

Figure 41: Figure shows the profile screen when it is updated

The figure illustrates the updated profile screen after the user's profile information has been successfully modified. It shows the latest details entered by the user, indicating that the system has saved and displayed the updated information correctly. This allows the user to verify that their profile changes have been applied successfully.

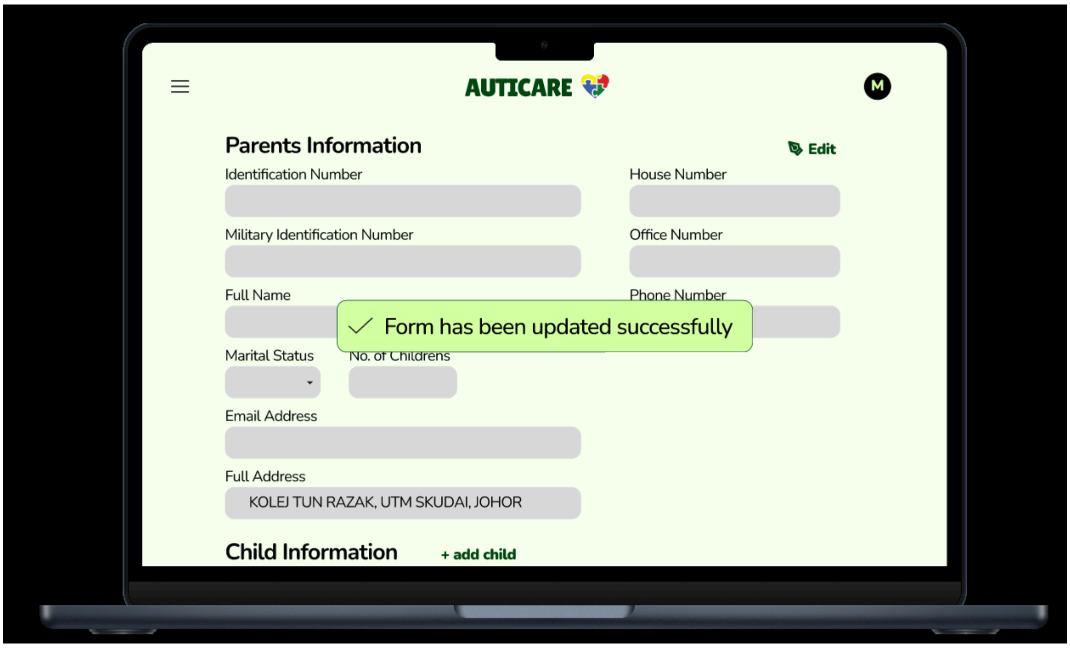

Figure 42: Figure shows that the form has been updated successfully

The figure illustrates the successful update of the form after the user's changes have been saved. It shows that the system has processed the user's input and updated the information accordingly. A confirmation message is displayed to notify the user that the update operation has been completed successfully.

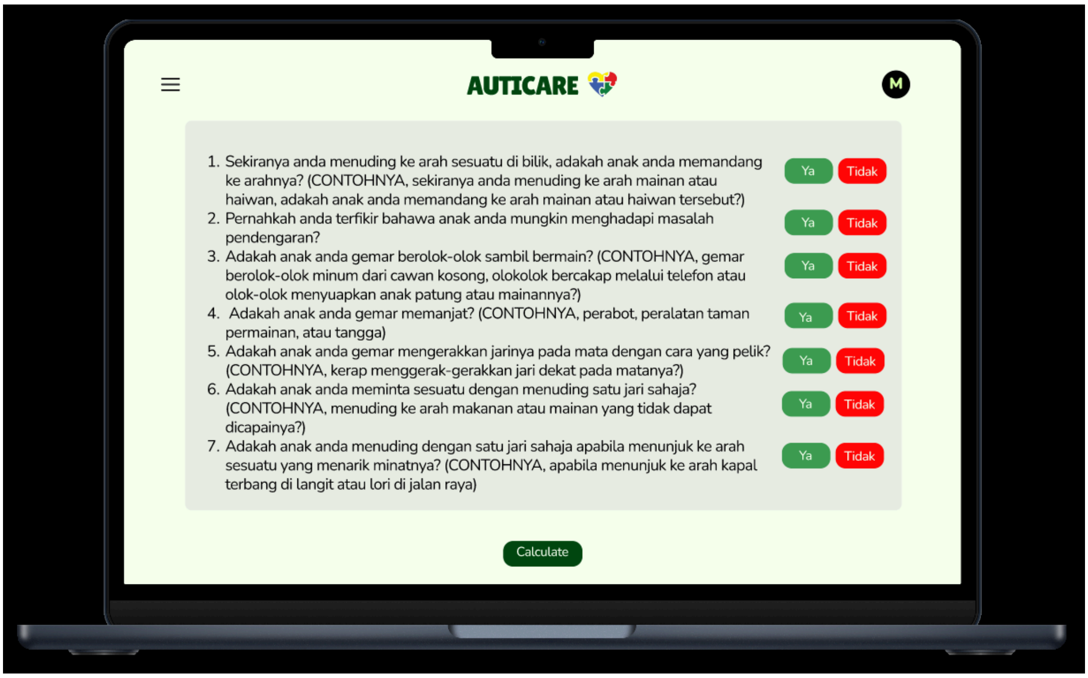

Figure 43: Figure shows the M-CHAT-R assessment

The figure illustrates the M-CHAT-R assessment screen, which displays a series of screening questions used to evaluate the possibility of autism spectrum disorder (ASD) in young children. Users can answer the questions based on the child's behaviour and development, and the system uses the responses to generate an assessment result for further understanding and guidance.

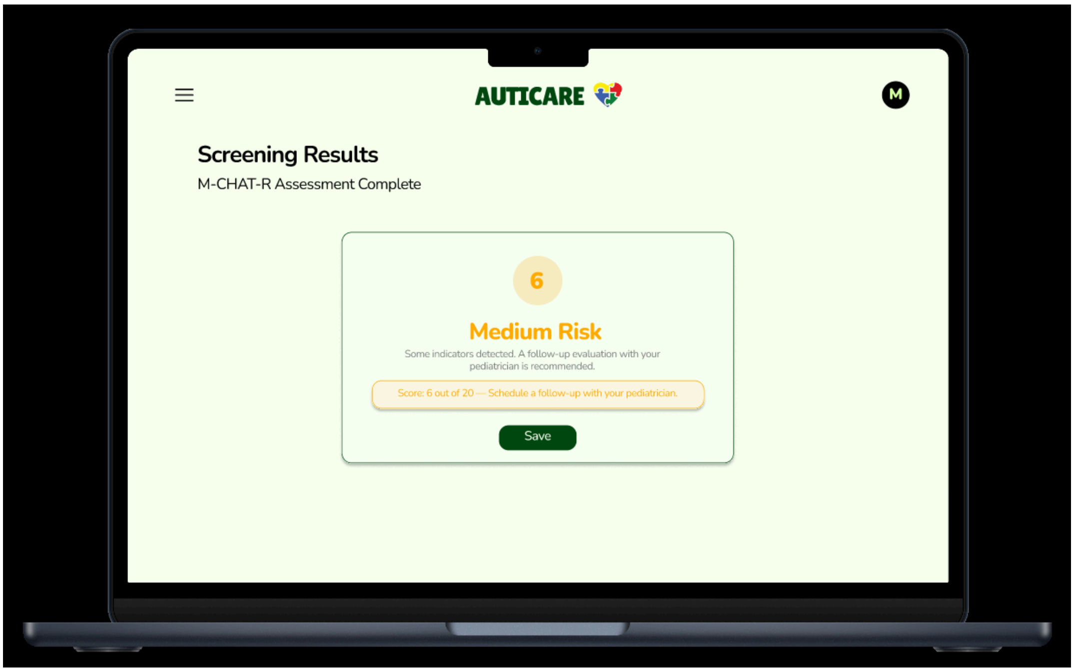

Figure 44: Figure shows the screening results

The figure illustrates the screening results generated after completing the M-CHAT-R assessment. It displays the outcome of the assessment based on the user's responses and provides an indication of the child's screening score and risk level. These results help parents understand the screening outcome and determine whether further professional evaluation may be recommended.

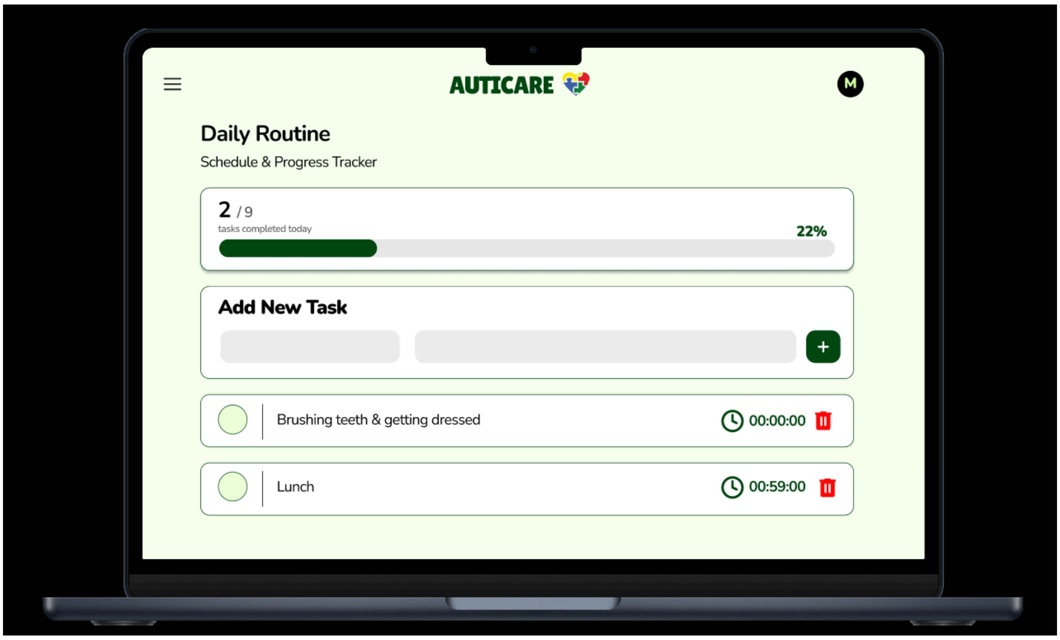

Figure 45: Figure shows the list of daily routine

The figure illustrates the list of daily routines available in the system. It displays the scheduled activities that have been created to help parents organize and monitor the child's daily tasks. This feature allows users to keep track of routines such as meals, hygiene, and sleep routine to support better daily management.

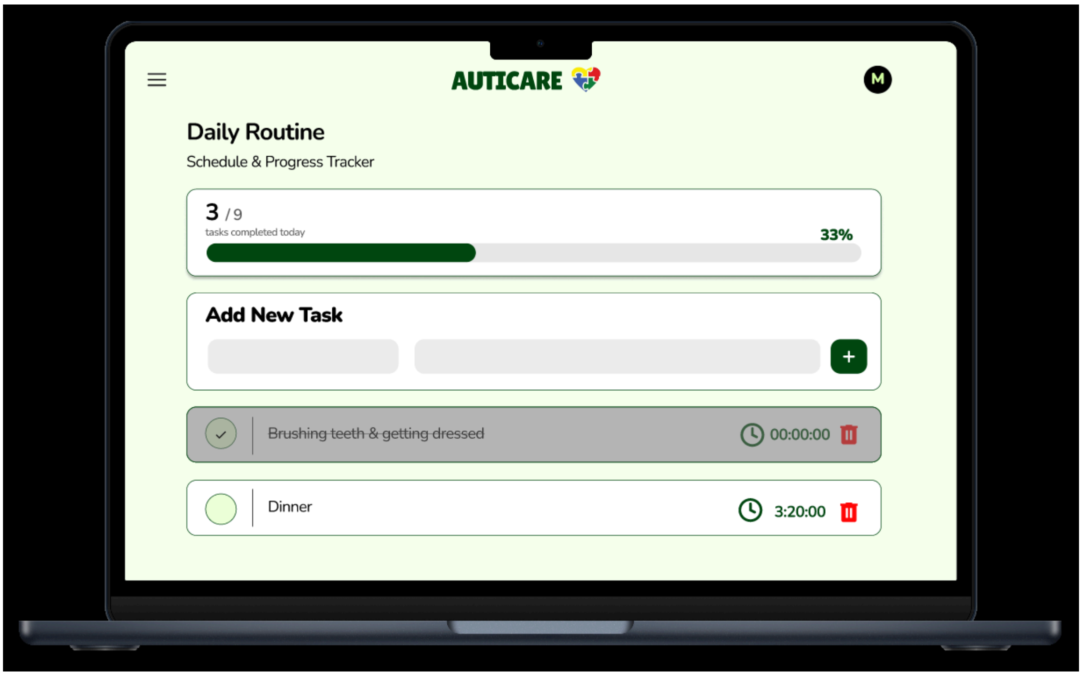

Figure 46: Figure shows that lunch has been deleted

The figure illustrates the successful deletion of the lunch activity from the daily routine list. It shows that the selected routine item has been removed from the system after the user confirms the deletion action. This allows users to manage and update the child's daily schedule by removing activities that are no longer needed.

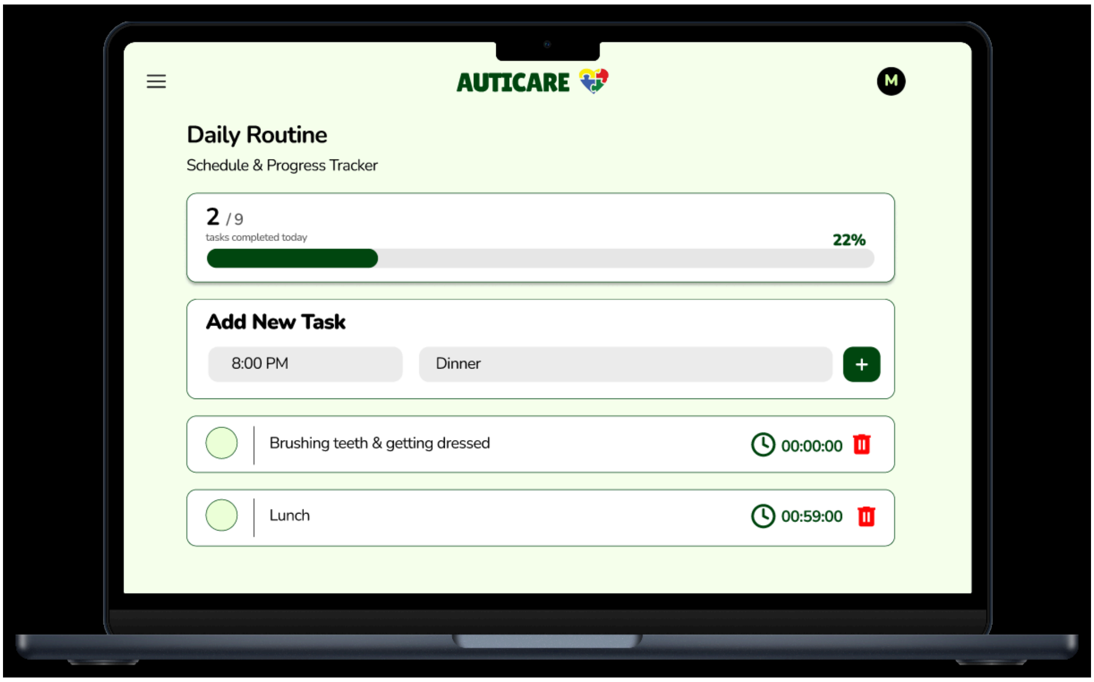

Figure 47: Figure shows that the user wants to enter dinner as the new task

The figure illustrates the process where the user enters "Dinner" as a new task in the daily routine section. It shows the input field being used to add a new activity into the system. This feature allows parents to create and customize daily routines based on the child's schedule and needs.

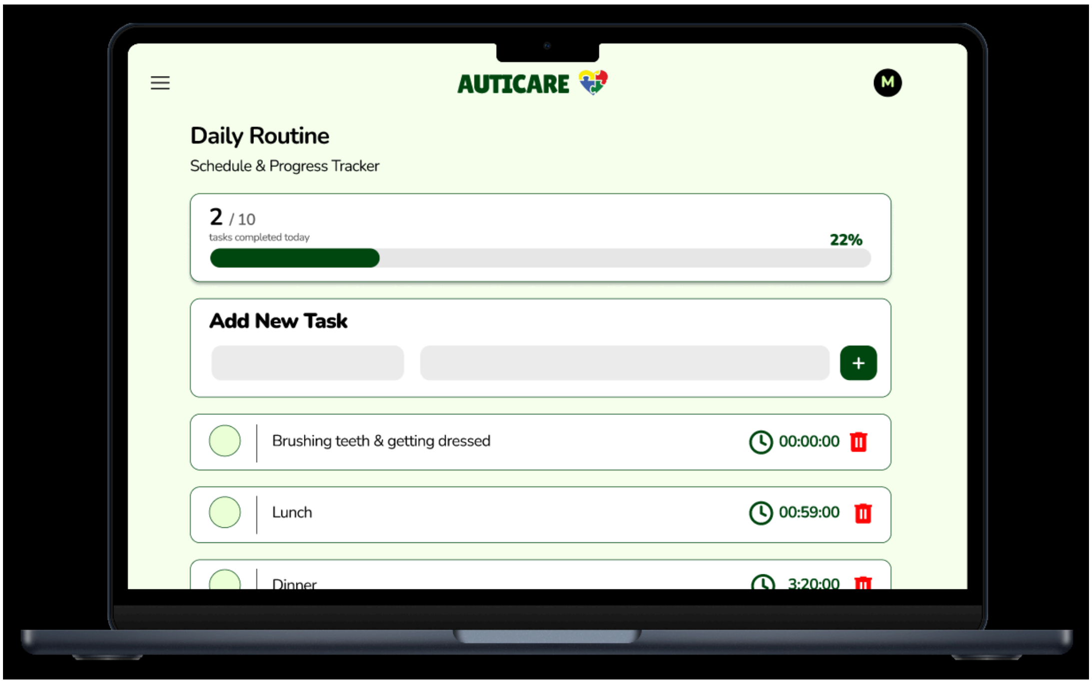

Figure 48: Figure shows that the new task has been added successfully

The figure illustrates that the new task has been added successfully to the daily routine list. It shows that the "Dinner" activity has been saved and displayed in the system, confirming that the task creation process has been completed. This feature allows users to update and maintain the child's daily schedule by adding new activities when needed.

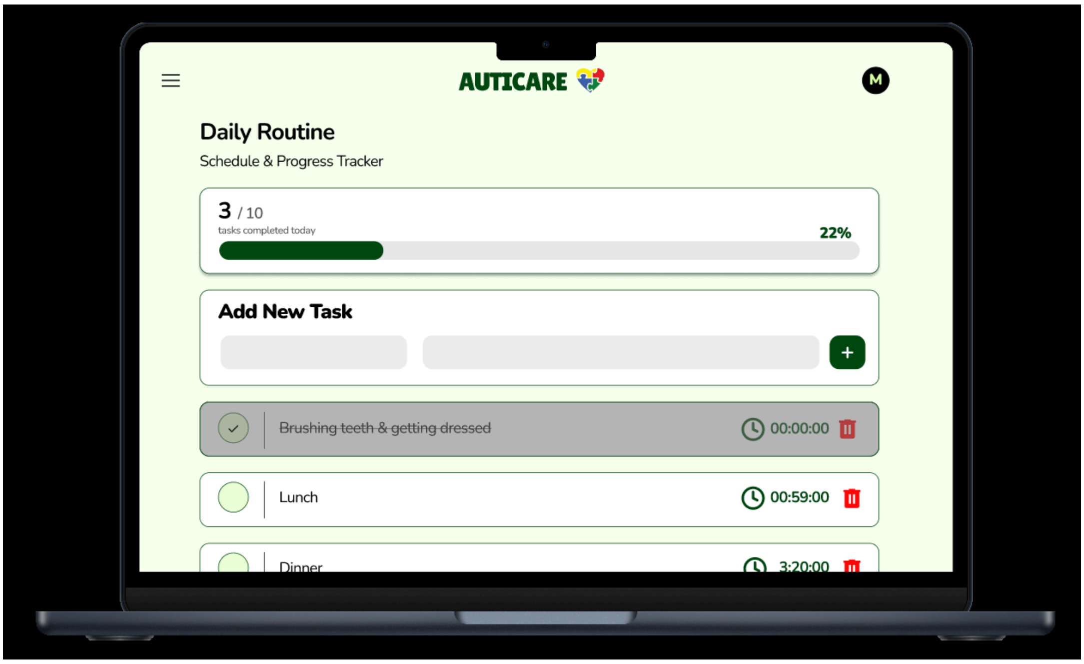

Figure 49: Figure shows that the task has been completed

This figure displays the confirmation status after the user marks the selected task as completed. The system updates the task status to indicate completion, allowing users to track their daily routine progress and easily identify which activities have already been accomplished.

Figure 50: Figure shows the screening summaries for each child

This figure displays the screening summary records for each child, providing an overview of their assessment results and screening score history. The information allows users, such as parents or caregivers, to review each child's screening progress and monitor the results over time.

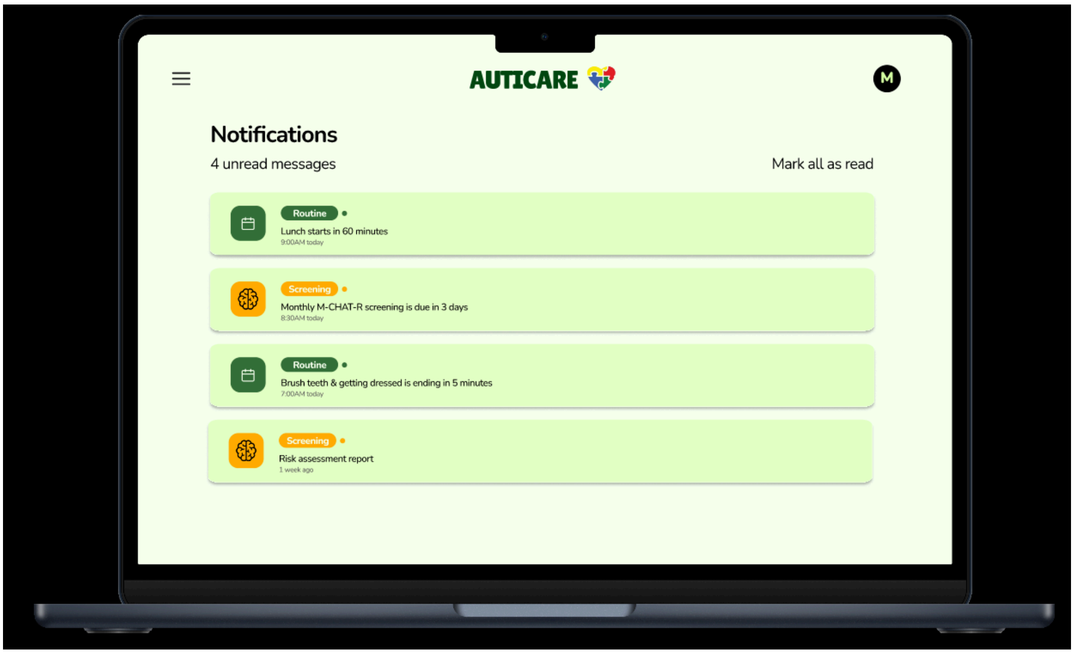

Figure 51: Figure shows the list of message received in the inbox

This figure displays the collection of messages stored in the user's inbox, allowing users to view notifications, updates, or communication records from the system. The inbox feature helps users keep track of important information and access previous messages whenever needed.

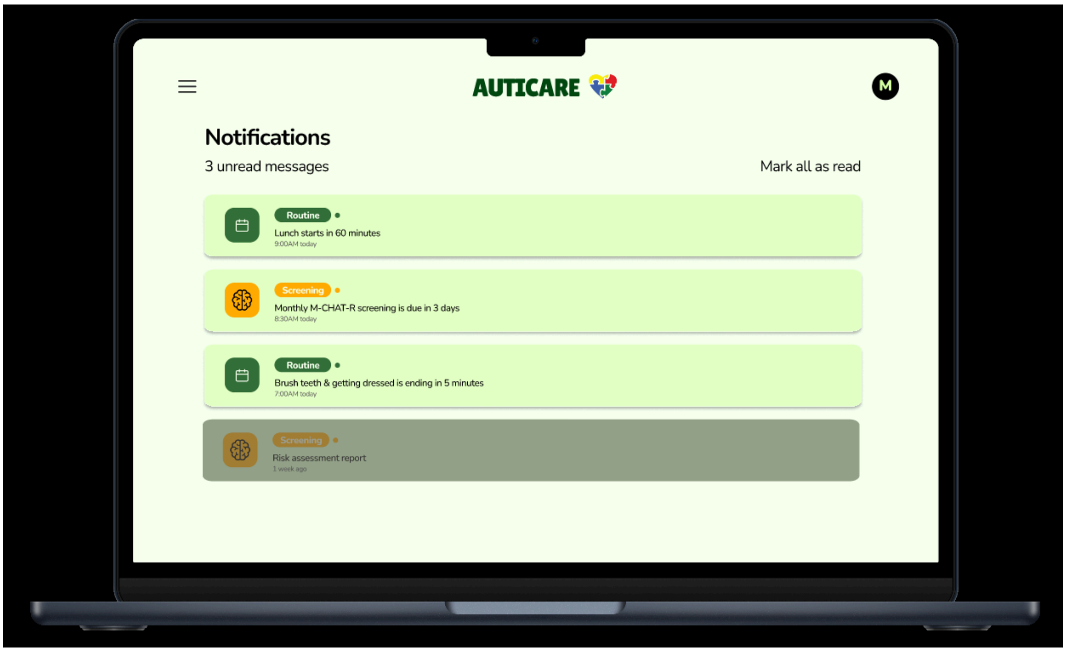

Figure 52: Figure shows the read message turns faded

This figure displays the inbox after a message has been opened and read by the user. The system changes the appearance of the read message by making it faded, allowing users to easily differentiate between unread and previously viewed messages. This helps users manage their messages more efficiently.

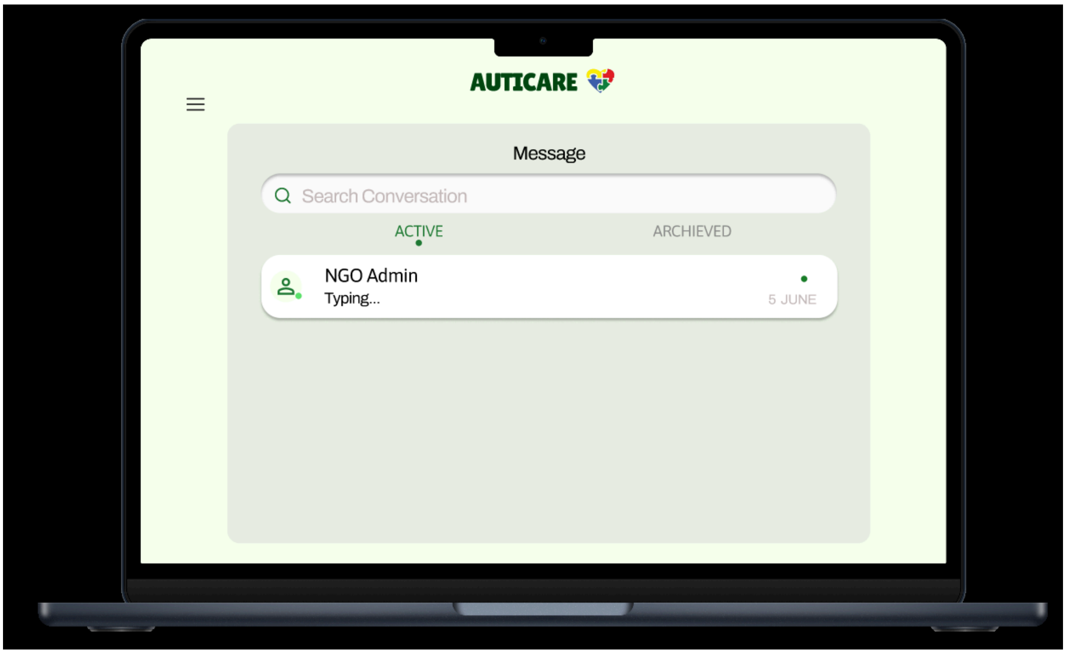

Figure 53

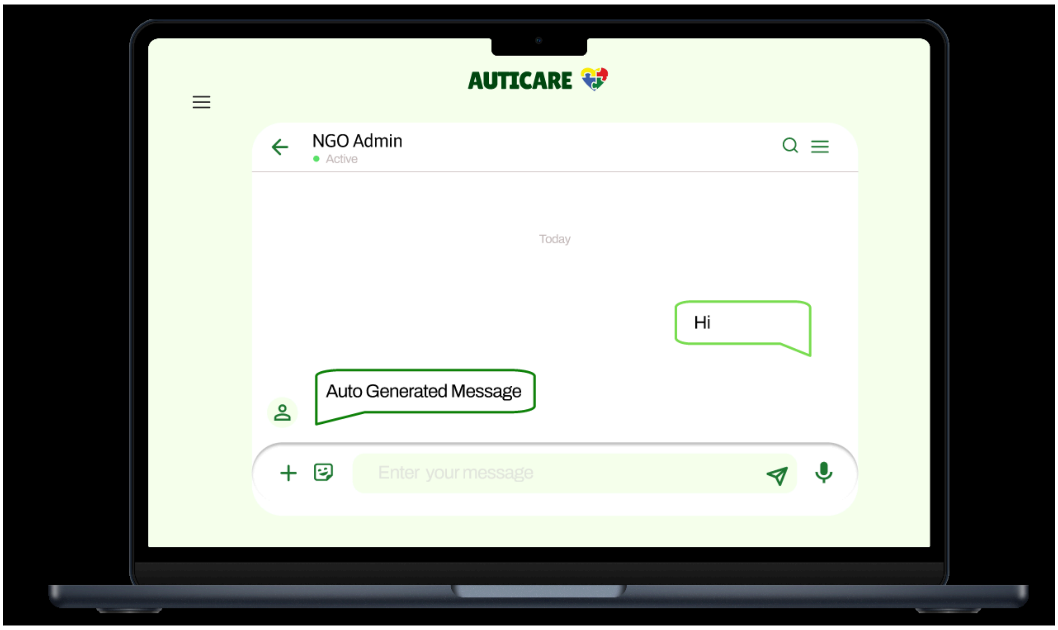

Figure 54: Figure shows message between user and NGO

Figure 53 and Figure 54 displays the communication interface between the user and the NGO, allowing both parties to exchange messages and share relevant information. The messaging feature enables users to communicate directly with the NGO for support, enquiries, or further assistance related to the system.

Link for reference:
https://www.figma.com/proto/h2qWaHSHZgwjnZPZdquzkb/SAD?node-id=41-267&p=f&t=6jaO2kLLrflS6VqC-1&scaling=scale-down&content-scaling=fixed&page-id=4%3A930

9.0 Summary of the Proposed System

The proposed system, AUTICARE, is a web-based Autism Support System designed to assist parents in supporting individuals with Autism Spectrum Disorder (ASD). The system integrates early ASD screening, personal information management, daily routine monitoring and progress analysis within a single platform to provide a more structured and accessible support environment.

The User Registration and Authentication allow users to securely create accounts and access the system. After registration, users can complete and manage their personal profiles through the Profile Management Module, which stores important background information required for personalized support.

The ASD Early Screening Module provides parents with the M-CHAT-R screening tool to help identify potential early signs of ASD. The application analyzes the outcome and sorts the degree of risks involved to enable users to comprehend the screening results easily.

The Daily Routine Tracker Module assists ASD children in developing daily routines by providing their parents with tools to track their activities such as eating, sleep, and personal hygiene habits. This module is geared towards fostering independent activity routine management for such individuals.

Notification features allow users to be notified of all scheduled activities or any updates concerning important routine actions. This way, the possibility of forgetting to undertake activities will be reduced significantly.

Data analysis can be done through the Reports and Analytics Module to assist users in making well-informed decisions about their care. The tool gathers all structured data obtained during routine activities and screening procedures for the generation of reports and visualization of progress.

Generally, the application ensures the integration of ASD screening, routine activities management, and reporting within a single platform. Through this platform, information exchange and sharing between healthcare providers and parents will be easier.

10.0 References

Prototype Demontration Video link: https://drive.google.com/file/d/1iB_EBI-uJOIoQEPzy21FFvI2ehqdDfI6/view?usp=drivesdk

11.0 Appendix

- Meeting 1

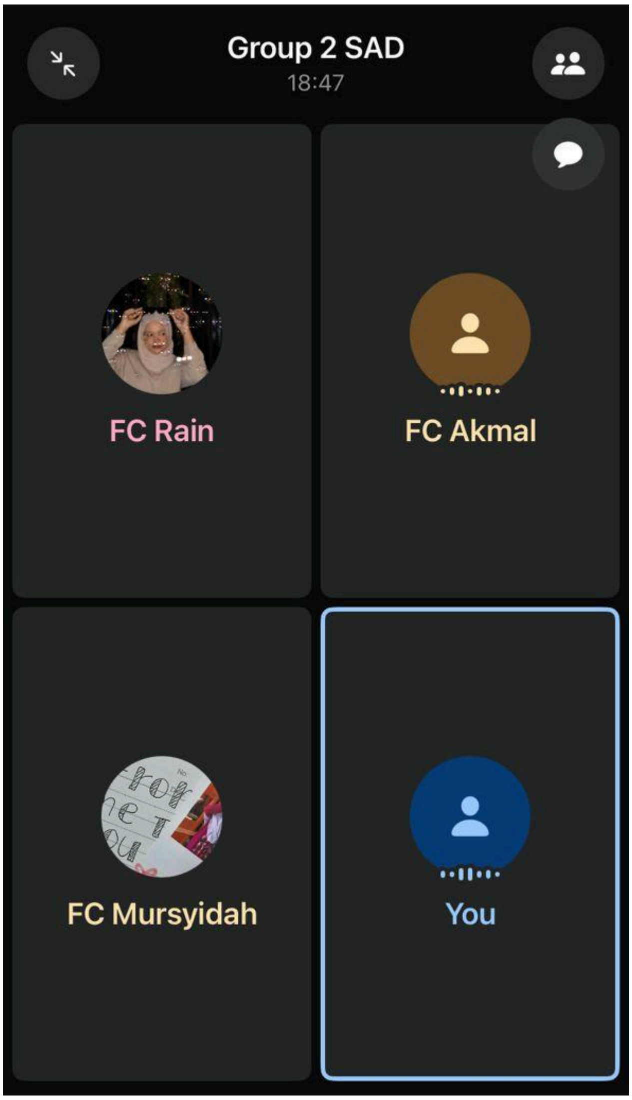

Figure 55: Figure shows the proof for meeting 1

● Meeting 2

Figure 56: Figure shows the proof for meeting 2

● Meeting 3

Figure 57: Figure shows the proof for meeting 3